

# Action-Objective Regret for Fixed-Action Adaptive Inference

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## Abstract

Adaptive inference systems spend different amounts of computation on different inputs, but the rule that allocates this computation is usually learned or calibrated from imperfect signals. This paper studies the deployment risk of that allocation rule after a set of compute actions has been fixed. For each action  $a$ , we write the conditional action objective as  $q_a(x) = r_a(x) + \lambda c_a(x)$ , combining predictive loss and inference cost, and view adaptive inference as choosing the action with the smallest estimated objective.

Within this fixed-action abstraction, the central organizing result is an oracle-regret bound for plug-in compute allocation. If the estimated action objectives are uniformly within  $\eta$  of the true objectives, the selected policy has at most  $2\eta$  excess cost-augmented risk; this constant is worst-case sharp. A margin-localized refinement, including a finite-sample binned version, shows that allocation errors matter only near action-tie boundaries, yielding faster rates when such near-ties are rare. This makes difficulty calibration an action-ordering problem rather than a generic claim that cost regularization is new.

We then instantiate finite-sample calibration guarantees for finite policy classes, binned difficulty policies, Monte Carlo action evaluation, ordinary logged feedback, and nested sequential logs in which deeper computation reveals prefix-action outcomes. Lower bounds show that weak logged overlap can be an information barrier, not just an artifact of IPS analysis, and that terminal-only logs can be parametrically less informative than nested prefix logs from the same runs. The same action-value view yields a Lagrangian interpretation of average compute budgets, a marginal-utility stopping rule for ordered computation, and simple ceilings for verifier-guided and latent-difficulty resampling. Reproducible synthetic, text-classifier routing, and handwritten-digits checks instantiate the assumptions and illustrate the predicted failure modes; they are CPU-scale sanity checks for the theory, not large-scale LLM routing benchmarks.

## 1 Introduction

Many contemporary AI systems do not use a fixed amount of computation for every input. A classifier may exit before all layers are evaluated, a transformer may route some tokens through deeper computation, and a language model may sample or verify many candidate solutions for hard problems. This shift makes inference compute a decision variable rather than a constant property of a model.

The usual empirical question is whether extra computation improves accuracy, reward, or benchmark score. A complementary theoretical question is more basic: who decides where the extra computation is spent, and what happens when that decision rule is wrong? In an adaptive inference system, the deployment policy must estimate whether an input is easy, whether another computation step has positive marginal value, or whether a generated answer should be refined. These estimates are learned, calibrated, or otherwise inferred from finite data.

This paper develops a small theory around that decision. We model inference as a choice among compute actions. Each action has a predictive loss and an inference cost. A compute policy maps each input to an action, and its deployment quality is measured by a cost-augmented risk. The central object is therefore not only a predictor but a policy for allocating computation.

Our contribution is a unified decision-theoretic account of fixed-action adaptive inference: once the available compute actions are fixed, deployment quality is controlled by how accurately the system estimates the cost-augmented action objectives  $q_a(x) = r_a(x) + \lambda c_a(x)$ . Concretely, we show:

- a compact cost-augmented policy formulation for adaptive inference, intended as common notation rather than a claim that cost-sensitive learning is new;
- a plug-in allocation theorem: uniform objective error  $\eta$  implies at most  $2\eta$  excess cost-augmented risk, with a sharpness example, an expected-error version, a margin-localized refinement, and a finite-sample bin-margin version for fixed difficulty bins;
- finite-sample calibration guarantees for finite policy classes, fixed binned difficulty policies, finite learned-binning classes, Monte Carlo action evaluation, known-propensity logged feedback, nested sequential logs, and clipped IPS with explicit bias, together with feedback lower bounds separating terminal-only from prefix-observing logs;
- links from the same action-objective view to average compute budgets, marginal-utility stopping for ordered computation, verifier-guided resampling limits, and latent-difficulty ceilings for best-of- $k$ ;
- reproducible PC-scale checks that instantiate the assumptions and expose their failure modes on synthetic allocation, Monte Carlo evaluation, logged and clipped off-policy feedback, verifier and latent-difficulty resampling, text-classifier routing, and handwritten digits.

The results are deliberately modest. We do not claim that compute regularization is new, nor that existing learning theory ignores computation. Instead, the goal is to isolate a specific deployment-time question that appears across modern adaptive inference: how errors in difficulty or utility estimation translate into errors in compute allocation.

The paper proceeds as follows. After related work and setup, we first analyze plug-in allocation from estimated action objectives. We then give finite-sample calibration guarantees under several feedback models, connect the objective to budgeted and ordered computation, and study two simple repeated-sampling ceilings. The final empirical section checks these mechanisms on PC-scale synthetic and digits experiments.

## 2 Related Work

Training scaling laws connect model size, data, and training compute to loss [1, 2]. Recent work extends the discussion toward inference or test-time compute, including repeated sampling, verification, search, and compute-optimal allocation [3, 4, 5, 6, 7, 8, 9, 10, 11]. A concurrent survey

distinguishes fixed compute controllability from input-adaptive test-time compute [12]. Our paper does not compete with these empirical or algorithmic systems; it isolates the error propagation from estimated action values to deployment risk.

LLM test-time compute is often implemented by self-consistency, verifiers, search, cascades, or learned routing [13, 14, 15, 16, 17, 18, 19, 20, 21, 22]. Imperfect verifiers can also limit inference scaling by admitting false positives [23]. Learned routing systems such as RouteLLM and Routoo choose among stronger and weaker language models using preference data, performance predictors, or cost-aware selectors [19, 20]. Other input-adaptive methods predict reward distributions, compute utility, or budget allocations for reasoning workloads [21, 10, 11]. Very recent work further studies constrained allocation policies, bandit-style allocation, training-free difficulty proxies, introspective reward-cost signals, evolving in-context demonstrations, and overthinking under long reasoning budgets [9, 11, 24, 25, 26, 27]. These works are close in motivation, but they primarily propose algorithms, proxy signals, or empirical allocation tradeoffs. Our contribution is narrower and more theoretical: we treat the available actions as fixed and analyze regret induced by imperfect estimates of the cost-augmented action curve. When one of these systems supplies a difficulty, reward, or cost predictor, our results can be read as a diagnostic layer for how estimation error and feedback limitations affect the resulting allocation policy. In particular, this paper should not be read as introducing input-adaptive test-time compute allocation. Existing systems already learn when to route, sample, continue computation, or stop. The narrower claim is that, once a candidate action set is fixed, many such systems share the same statistical bottleneck: estimating the conditional action objectives accurately enough that plug-in allocation has small deployment regret.

Table 1 summarizes the nearest boundaries. The table is not meant to rank prior work; it makes explicit which parts of the surrounding literature this paper reuses and which narrower question it analyzes.

Adaptive computation has a longer history in neural networks, including adaptive computation time [28], early exits [29, 30, 31, 32], dynamic routing and sparse computation [33, 34, 35, 36, 37], and recent PAC-Bayes analyses of adaptive depth [38]. Ordered compute actions in this paper cover early exit and repeated sampling as special cases, but the diminishing-return assumption is not intended to hold universally.

## Nearest Prior Work and Novelty Boundary

Classical budgeted prediction, cascades, and learning with costs are also directly relevant [39, 40, 41, 42, 43, 44]. These works already study accuracy-cost tradeoffs and input-dependent acquisition or classifier selection. The present paper should therefore be read as a compact abstraction specialized to modern adaptive inference, not as the invention of cost-sensitive learning. Compared with budgeted prediction and cascade learning, our model removes much of the sequential design problem: the action set is assumed fixed, and the paper does not learn feature-acquisition policies or cascade architectures. Its role is instead to expose the calibration problem that remains after such actions have been chosen: estimating the conditional cost-augmented objective well enough to select the right action at deployment time.

Our formalism also touches structural risk minimization, minimum description length, PAC-Bayes, plug-in classification, policy learning, and metareasoning [45, 46, 47, 48, 49, 50, 51]. The finite-class and PAC-Bayes results below are direct applications of standard tools to a deployment loss that includes inference cost. The more specific focus is the conditional action objective used by an adaptive inference policy, and the regret caused by estimating that objective imperfectly. The logged-feedback results should also be read through the lens of policy learning and off-policy

Table 1: Nearest prior work and the role of the present paper.

| Line of work                              | What is already covered                                                                                     | Role of this paper                                                                                                          |
|-------------------------------------------|-------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------|
| Test-time compute allocation and routing  | Algorithms and empirical tradeoffs for routing, sampling, stopping, or allocating reasoning budgets.        | Takes the deployable actions as fixed and analyzes how action-objective estimation error propagates to allocation regret.   |
| Budgeted prediction and cascades          | Accuracy-cost objectives, sequential feature acquisition, classifier cascades, and budgeted expert advice.  | Removes the architecture/acquisition design problem and studies the residual calibration problem for a fixed action menu.   |
| Policy learning and off-policy evaluation | Uniform policy-value bounds, inverse-propensity estimators, clipping, and observational-data assumptions.   | Instantiates these tools for compute-policy selection with cost-augmented action losses.                                    |
| Plug-in classification and margin theory  | Stability and fast rates when class posteriors are estimated accurately and decision margins are favorable. | Applies the margin view to action objectives whose coordinates are inference-time compute actions rather than class labels. |
| Metareasoning and adaptive depth          | Computation as an action with value and cost; models that adapt depth or computation dynamically.           | Gives finite-sample regret statements for fixed inference actions when the value-cost curve must be calibrated from data.   |

evaluation, including inverse-propensity, doubly robust, and counterfactual-risk tools for logged terminal actions [52, 53, 54, 50]. Bandits with side observations and feedback graphs study actions that reveal other actions’ losses [55, 56]; inverse-probability weighting for coarsened data is classical [57, 58]. We do not introduce a new IPS, doubly robust, IPCW, clipping, or graph-feedback estimator, nor do we address nuisance estimation, semiparametric efficiency, hidden confounding, or adaptive logging. The point is to instantiate these standard estimators for compute-policy selection, where the policy action is an inference-time compute choice and the outcome is the cost-augmented action loss. In this view, prefix-observing logs are a deterministic side-observation or coarsening pattern whose observation probability for action  $k$  is survival to that prefix, not exact terminal stopping there. The nested-log result below is related to per-decision importance sampling in sequential off-policy evaluation [59], but specializes the idea to ordered compute actions where deeper execution reveals prefix-action outcomes. Classical metareasoning already treats computation as an action with value and cost [51]; here we specialize that principle to fixed inference actions and give simple statistical regret statements for the case where the value curve must be learned or calibrated from finite data.

### 3 Setup

Let  $(X, Y) \sim P$  take values in  $\mathcal{X} \times \mathcal{Y}$ . An adaptive inference system has a finite set of compute actions  $\mathcal{A}$ . An action may represent exiting at a given depth, using a larger subnetwork, sampling  $k$  answers, running a verifier, or taking one more reasoning step.

For each action  $a \in \mathcal{A}$ , let  $\ell_a(x, y) \in [0, 1]$  be the conditional expected loss incurred when action  $a$  is used on input  $x$  and label  $y$ , averaging over any internal generation or routing randomness. Let  $c_a(x) \in [0, C]$  be the conditional expected inference cost. The cost may measure wall-clock latency,

floating-point operations, token budget, monetary cost, or a normalized proxy.

**Standing assumptions.** We assume  $\mathcal{X}$  and  $\mathcal{Y}$  are standard Borel spaces,  $\mathcal{A}$  is finite, and all policies, losses, and costs are measurable. The policy class, action losses, and costs are fixed independently of the evaluation sample; equivalently, they may be trained on separate data and then evaluated on a fresh calibration or validation sample. Unless stated otherwise, empirical risks below are full-information quantities: for every validation example, the analyst can evaluate the bounded loss and cost of every action. If these quantities are obtained through Monte Carlo estimates, Corollary 5 adds a concentration term for the estimation layer. If only logged actions are observed, Corollary 6 and Corollary 8 replace full-information calibration with known-propensity off-policy estimates under support assumptions.

**Definition 1** (Compute policy). *A compute policy is a measurable function  $\pi : \mathcal{X} \rightarrow \mathcal{A}$ . The policy class is denoted by  $\Pi$ .*

**Definition 2** (Cost-augmented risk). *For tradeoff parameter  $\lambda \geq 0$ , the cost-augmented risk of policy  $\pi$  is*

$$J_\lambda(\pi) = \mathbb{E} [\ell_{\pi(X)}(X, Y) + \lambda c_{\pi(X)}(X)] . \quad (1)$$

*Given a sample  $S = \{(x_i, y_i)\}_{i=1}^n$ , its empirical counterpart is*

$$\hat{J}_{\lambda, S}(\pi) = \frac{1}{n} \sum_{i=1}^n [\ell_{\pi(x_i)}(x_i, y_i) + \lambda c_{\pi(x_i)}(x_i)] . \quad (2)$$

The parameter  $\lambda$  sets the exchange rate between predictive loss and inference cost. A high  $\lambda$  favors cheap actions; a low  $\lambda$  favors accuracy regardless of cost. Throughout, write  $B = 1 + \lambda C$  for the common range bound on the cost-augmented objective.

## 4 Difficulty-Calibrated Allocation

We first isolate the pointwise allocation problem. Many adaptive systems estimate a difficulty or utility curve for each input and choose the action with the best predicted tradeoff. The question is how errors in that estimated curve translate into deployment regret.

Define the conditional risk curve using a regular conditional distribution:

$$r_a(x) = \int \ell_a(x, y) P(dy \mid x) . \quad (3)$$

Let the true cost-augmented action objective be

$$q_a(x) = r_a(x) + \lambda c_a(x) . \quad (4)$$

For any measurable compute policy  $\pi$ ,

$$J_\lambda(\pi) = \mathbb{E} [q_{\pi(X)}(X)] , \quad (5)$$

by the tower property. The oracle cost-augmented action is

$$a_\lambda^*(x) \in \arg \min_{a \in \mathcal{A}} q_a(x) . \quad (6)$$

Let  $\pi_\lambda^*(x) = a_\lambda^*(x)$ .

In practice, the system uses a measurable estimated objective  $\hat{q}_a(x) = \hat{r}_a(x) + \lambda \hat{c}_a(x)$  and chooses

$$\hat{\pi}_\lambda(x) \in \arg \min_{a \in \mathcal{A}} \hat{q}_a(x). \quad (7)$$

All argmins use a fixed measurable tie-breaking rule. If the estimates are learned from training or calibration data, the statements below are understood conditioned on that data; the displayed objective-error events are over the fresh deployment draw  $X$ .

**Theorem 1** (Regret from objective estimation error). *Suppose that  $P_X$ -almost surely*

$$\max_{a \in \mathcal{A}} |\hat{q}_a(X) - q_a(X)| \leq \eta. \quad (8)$$

*Then*

$$J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq 2\eta. \quad (9)$$

*Proof.* Fix an  $x$  where the objective-error bound holds. Since  $\hat{\pi}_\lambda(x)$  minimizes  $\hat{q}_a(x)$ ,

$$q_{\hat{\pi}_\lambda(x)}(x) \leq \hat{q}_{\hat{\pi}_\lambda(x)}(x) + \eta \quad (10)$$

$$\leq \hat{q}_{\pi_\lambda^*(x)}(x) + \eta \quad (11)$$

$$\leq q_{\pi_\lambda^*(x)}(x) + 2\eta. \quad (12)$$

Taking expectation over  $X$  proves the claim.  $\square$

**Corollary 1** (Risk-only and cost-estimation special cases). *If costs are known and  $\max_a |\hat{r}_a(X) - r_a(X)| \leq \epsilon$   $P_X$ -almost surely, then  $J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq 2\epsilon$ . If both risks and costs are estimated with errors  $\epsilon_r$  and  $\epsilon_c$ , then the same conclusion holds with  $\eta = \epsilon_r + \lambda \epsilon_c$ .*

**Theorem 2** (Margin-localized allocation regret). *For a fixed  $x$ , let  $e(x) = \max_a |\hat{q}_a(x) - q_a(x)|$  and let  $q_{(1)}(x) = \min_{a \in \mathcal{A}} q_a(x)$ . Define the action margin by*

$$\gamma(x) = \inf\{q_a(x) - q_{(1)}(x) : q_a(x) > q_{(1)}(x), a \in \mathcal{A}\}, \quad (13)$$

*with the convention  $\inf \emptyset = \infty$ . Then*

$$q_{\hat{\pi}_\lambda(x)}(x) - q_{\pi_\lambda^*(x)}(x) \leq 2e(x) \mathbf{1}\{\gamma(x) \leq 2e(x)\}. \quad (14)$$

*Consequently, if  $e(X) \leq \epsilon$   $P_X$ -almost surely, then*

$$J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq 2\epsilon P_X(\gamma(X) \leq 2\epsilon). \quad (15)$$

*Proof.* Theorem 1 gives the pointwise bound  $q_{\hat{\pi}_\lambda(x)}(x) - q_{\pi_\lambda^*(x)}(x) \leq 2e(x)$ . If  $\gamma(x) > 2e(x)$ , no nonoptimal action can have estimated objective as small as a true minimizer, so any estimated minimizer is also a true minimizer and the pointwise regret is zero. Combining the two cases proves the claim.  $\square$

These results identify a target for learning a difficulty estimator. The estimator need not directly predict correctness; it must predict the cost-augmented ordering of available compute actions accurately enough. The margin-localized bound explains why ambiguous inputs dominate allocation errors: when two compute actions are nearly tied, small estimation errors can change the selected action. This is the same kind of plug-in stability phenomenon that appears in classification margin analyses [49]. The distinction here is not the argmin argument itself, but the object to which it is applied: a conditional action objective whose coordinates are deployable inference actions and whose errors include both predictive loss and inference cost. Thus the margin is an action-allocation margin, not a class-label margin.

**Corollary 2** (Expected and localized objective error). *With  $e(x) = \max_a |\hat{q}_a(x) - q_a(x)|$ ,*

$$J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq 2\mathbb{E}[e(X)]. \quad (16)$$

*If the margin  $\gamma(x)$  is defined as in Theorem 2, then the localized refinement*

$$J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq 2\mathbb{E}[e(X)\mathbf{1}\{\gamma(X) \leq 2e(X)\}] \quad (17)$$

*also holds.*

*Proof.* The proof of Theorem 1 gives the pointwise inequality  $q_{\hat{\pi}_\lambda(x)}(x) - q_{\pi_\lambda^*(x)}(x) \leq 2e(x)$  wherever the objectives are defined. The proof of Theorem 2 gives the stronger pointwise localized inequality. Taking expectations gives both claims.  $\square$

**Corollary 3** (Margin-tail fast rate). *Assume the conditions of Theorem 2 and suppose that for some constants  $A \geq 0$  and  $\kappa \geq 0$ ,*

$$P_X(\gamma(X) \leq t) \leq At^\kappa \quad \text{for all } t \geq 0. \quad (18)$$

*If  $e(X) \leq \epsilon$   $P_X$ -almost surely, then*

$$J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq A2^{\kappa+1}\epsilon^{\kappa+1}. \quad (19)$$

*More generally, the right-hand side may be replaced by  $2\epsilon \min\{1, A(2\epsilon)^\kappa\}$  when the margin-tail condition is known only locally or the displayed bound exceeds  $2\epsilon$ .*

*Proof.* Theorem 2 gives  $J_\lambda(\hat{\pi}_\lambda) - J_\lambda(\pi_\lambda^*) \leq 2\epsilon P_X(\gamma(X) \leq 2\epsilon)$ . Applying the margin-tail condition gives the first inequality. The capped version also uses the trivial bound  $P_X(\gamma(X) \leq 2\epsilon) \leq 1$ .  $\square$

**Proposition 1** (Sharpness of the factor two). *The constant 2 in Theorem 1 cannot be improved under only the uniform objective-error assumption. More precisely, fix  $\eta > 0$  and  $\alpha \in (0, 1)$  such that  $(2 - \alpha)\eta \leq 1$ . There is a one-point distribution, two compute actions, zero costs, and losses in  $[0, 1]$  such that*

$$\max_{a \in \mathcal{A}} |\hat{q}_a(X) - q_a(X)| \leq \eta \quad (20)$$

*almost surely, but the plug-in minimizer of  $\hat{q}_a(X)$  is unique and has cost-augmented regret  $(2 - \alpha)\eta$ . Consequently, no uniform bound of the form  $c\eta$  with  $c < 2$  can hold over this class. If ties among estimated minimizers may be broken adversarially and  $2\eta \leq 1$ , the regret  $2\eta$  is attained exactly.*

*Proof.* Let  $X = x_0$  and  $Y = y_0$  almost surely, let  $\mathcal{A} = \{g, b\}$ , and set all costs to zero. Define the true losses by  $\ell_g(x_0, y_0) = 0$  and  $\ell_b(x_0, y_0) = (2 - \alpha)\eta$ , which are in  $[0, 1]$  by assumption. Thus  $q_g(x_0) = 0$  and  $q_b(x_0) = (2 - \alpha)\eta$ . Now set

$$\hat{q}_g(x_0) = \eta, \quad \hat{q}_b(x_0) = (1 - \alpha)\eta. \quad (21)$$

Both objective estimates have absolute error  $\eta$ , and the estimated objective strictly prefers action  $b$ . The oracle action is  $g$ , so the cost-augmented regret is  $q_b(x_0) - q_g(x_0) = (2 - \alpha)\eta$ . Since  $\alpha$  can be arbitrarily small, any constant smaller than two is impossible uniformly. When  $2\eta \leq 1$ , exact equality is obtained by setting  $q_b(x_0) = 2\eta$  and  $\hat{q}_g(x_0) = \hat{q}_b(x_0) = \eta$ , then break the estimated tie in favor of  $b$ .  $\square$

**Theorem 3** (Finite calibration for binned difficulty). *Let  $b : \mathcal{X} \rightarrow \{1, \dots, M\}$  be a fixed measurable binning of inputs, chosen independently of the calibration sample used below. Consider policies that choose a single action per bin. For bin  $m$ , define  $p_m = P_X(b(X) = m)$  and  $Q_{m,a} = \mathbb{E}[q_a(X) \mid b(X) = m]$ . Let  $\pi_b^*$  choose an action minimizing  $Q_{m,a}$  in each bin and let  $\hat{\pi}_b$  choose the action minimizing estimates  $\hat{Q}_{m,a}$ . If, for every bin with  $p_m > 0$ ,  $|\hat{Q}_{m,a} - Q_{m,a}| \leq \epsilon_m$  for all actions  $a$ , then*

$$J_\lambda(\hat{\pi}_b) - J_\lambda(\pi_b^*) \leq 2 \sum_{m=1}^M p_m \epsilon_m. \quad (22)$$

Moreover, with full-information calibration data, let  $n_m = |\{i : b(x_i) = m\}|$ . For nonempty bins define

$$\hat{Q}_{m,a} = \frac{1}{n_m} \sum_{i:b(x_i)=m} (\ell_a(x_i, y_i) + \lambda c_a(x_i)), \quad (23)$$

and set  $\hat{Q}_{m,a}$  arbitrarily when  $n_m = 0$ . Conditional on the bin counts, with probability at least  $1 - \delta$ ,

$$J_\lambda(\hat{\pi}_b) - J_\lambda(\pi_b^*) \leq 2 \sum_{m:n_m>0} p_m B \sqrt{\frac{\log(2M|\mathcal{A}|/\delta)}{2n_m}} + B \sum_{m:n_m=0} p_m. \quad (24)$$

*Proof.* Within each bin, the same argmin-stability proof as Theorem 1 gives at most  $2\epsilon_m$  excess objective relative to the best constant action in that bin. Averaging over bins gives the first inequality. For the finite-sample statement, conditional on the bin counts, Hoeffding's inequality applied to each nonempty bin-action objective and a union bound over at most  $M|\mathcal{A}|$  estimates give the displayed deviation simultaneously for all nonempty bins. The deterministic argument then controls those bins. On an empty bin the selected action can be arbitrary, but the excess objective is at most  $B$ , giving the additional  $B \sum_{m:n_m=0} p_m$  term.  $\square$

**Theorem 4** (Bin-margin-localized finite calibration). *In the setting of Theorem 3, define the best bin objective  $Q_{m,(1)} = \min_{a \in \mathcal{A}} Q_{m,a}$  and the bin action margin*

$$\Gamma_m = \inf\{Q_{m,a} - Q_{m,(1)} : Q_{m,a} > Q_{m,(1)}, a \in \mathcal{A}\}, \quad (25)$$

with the convention  $\inf \emptyset = \infty$ . If, for every bin with  $p_m > 0$ ,  $|\hat{Q}_{m,a} - Q_{m,a}| \leq \epsilon_m$  for all actions  $a$ , then

$$J_\lambda(\hat{\pi}_b) - J_\lambda(\pi_b^*) \leq 2 \sum_{m=1}^M p_m \epsilon_m \mathbf{1}\{\Gamma_m \leq 2\epsilon_m\}. \quad (26)$$

Moreover, with the full-information calibration estimator in Theorem 3, conditional on the bin counts, with probability at least  $1 - \delta$ ,

$$J_\lambda(\hat{\pi}_b) - J_\lambda(\pi_b^*) \leq 2 \sum_{m:n_m>0} p_m \epsilon_m(\delta) \mathbf{1}\{\Gamma_m \leq 2\epsilon_m(\delta)\} + B \sum_{m:n_m=0} p_m, \quad (27)$$

where

$$\epsilon_m(\delta) = B \sqrt{\frac{\log(2M|\mathcal{A}|/\delta)}{2n_m}}. \quad (28)$$

*Proof.* Fix a bin  $m$  where the deterministic error bound holds, and let  $\hat{a}_m$  be the action chosen by  $\hat{\pi}_b$  in that bin. The proof of Theorem 3 gives  $Q_{m,\hat{a}_m} - Q_{m,(1)} \leq 2\epsilon_m$ . If  $\hat{a}_m$  is a true minimizer, the bin contributes no excess objective. If it is not a true minimizer, then by definition  $Q_{m,\hat{a}_m} - Q_{m,(1)} \geq \Gamma_m$ ,



so this case can occur only when  $\Gamma_m \leq 2\epsilon_m$ . Multiplying the resulting binwise bound by  $p_m$  and summing over bins proves the deterministic statement.

For the finite-sample statement, use the same simultaneous Hoeffding event as in Theorem 3 on all nonempty bin-action pairs and set  $\epsilon_m = \epsilon_m(\delta)$ . Empty bins have arbitrary selected actions, and their excess objective is at most  $B$  because  $q_a(X) \in [0, B]$ .  $\square$

This localized binned result is the finite-sample bridge between difficulty calibration and allocation regret. It is not a new uniform-convergence theorem: it combines the binwise concentration argument above with the same argmin margin argument used in Theorem 2. Its localization is with respect to the averaged bin objectives  $Q_{m,a}$ , not the pointwise margin  $\gamma(X)$ . Calibration samples spent on bins whose best and second-best actions are well separated do not affect the regret bound once their confidence intervals are narrower than half the bin margin; the remaining statistical difficulty is concentrated on near-tie bins and on empty bins.

The independence condition is important. If a difficulty score or bin boundary is learned from the same calibration examples used to estimate  $Q_{m,a}$ , the same conclusion requires a uniform-convergence argument over that binning class, or a separate split for constructing the bins.

## 5 Calibration Data and Feedback Models

The previous section assumed an estimated action objective and analyzed the regret of the induced plug-in allocator. We now ask how calibration data can support this decision rule. The results in this section are standard finite-class concentration arguments applied to the cost-augmented deployment loss; their purpose is to make the required feedback model explicit.

**Theorem 5** (Finite policy class). *Assume  $\ell_a(x, y) \in [0, 1]$  and  $c_a(x) \in [0, C]$  for all  $a, x, y$ . Let  $B = 1 + \lambda C$ . For any finite policy class  $\Pi$  and any  $\delta \in (0, 1)$ , with probability at least  $1 - \delta$  over  $S \sim P^n$ , every  $\pi \in \Pi$  satisfies*

$$\left| J_\lambda(\pi) - \hat{J}_{\lambda,S}(\pi) \right| \leq B \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (29)$$

*Proof.* For a fixed  $\pi$ , the random variable  $Z_\pi = \ell_{\pi(X)}(X, Y) + \lambda c_{\pi(X)}(X)$  lies in  $[0, B]$ . Hoeffding's inequality bounds the two-sided deviation between its mean and empirical average. A union bound over  $\Pi$  completes the proof.  $\square$

**Corollary 4** (Compute-regularized empirical risk minimization). *Let  $\hat{\pi} \in \arg \min_{\pi \in \Pi} \hat{J}_{\lambda,S}(\pi)$ . Under the event of Theorem 5,*

$$J_\lambda(\hat{\pi}) \leq \inf_{\pi \in \Pi} J_\lambda(\pi) + 2B \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (30)$$

**Corollary 5** (ERM with Monte Carlo action-objective estimates). *Fix  $\lambda$  and a finite policy class  $\Pi$  independently of the calibration sample and Monte Carlo draws. Let  $z_a(x, y) = \ell_a(x, y) + \lambda c_a(x) \in [0, B]$ . For every calibration example  $(x_i, y_i)$  and action  $a$ , suppose we observe  $m$  conditionally independent random variables  $W_{i,a,1}, \dots, W_{i,a,m} \in [0, B]$  satisfying*

$$\mathbb{E}[W_{i,a,j} \mid S] = z_a(x_i, y_i), \quad (31)$$

*with independence across examples, actions, and Monte Carlo repetitions conditional on  $S$ . Define*

$$\tilde{J}_{\lambda,S,m}(\pi) = \frac{1}{n} \sum_{i=1}^n \frac{1}{m} \sum_{j=1}^m W_{i,\pi(x_i),j}. \quad (32)$$

Let  $\tilde{\pi} \in \arg \min_{\pi \in \Pi} \tilde{J}_{\lambda, S, m}(\pi)$ . Then, with probability at least  $1 - \delta$  over the calibration sample and the Monte Carlo draws,

$$J_{\lambda}(\tilde{\pi}) \leq \inf_{\pi \in \Pi} J_{\lambda}(\pi) + 2B \sqrt{\frac{\log(4|\Pi|/\delta)}{2n}} + 2B \sqrt{\frac{\log(4|\Pi|/\delta)}{2nm}}. \quad (33)$$

*Proof.* Theorem 5, applied with failure probability  $\delta/2$ , gives with probability at least  $1 - \delta/2$  that

$$\sup_{\pi \in \Pi} |J_{\lambda}(\pi) - \hat{J}_{\lambda, S}(\pi)| \leq B \sqrt{\frac{\log(4|\Pi|/\delta)}{2n}}. \quad (34)$$

Conditional on  $S$ , Hoeffding's inequality applied to the  $nm$  bounded Monte Carlo variables selected by a fixed policy  $\pi$  gives

$$P \left( \left| \tilde{J}_{\lambda, S, m}(\pi) - \hat{J}_{\lambda, S}(\pi) \right| > t \mid S \right) \leq 2 \exp(-2nmt^2/B^2). \quad (35)$$

A union bound over  $\Pi$  with failure probability  $\delta/2$  gives the second deviation term uniformly over policies. On the intersection of the two events,  $\sup_{\pi \in \Pi} |J_{\lambda}(\pi) - \tilde{J}_{\lambda, S, m}(\pi)|$  is at most the sum of the two displayed deviations. The usual ERM argument for  $\tilde{\pi}$  then doubles this uniform deviation.  $\square$

This corollary isolates the price of estimating action objectives by Monte Carlo. The first term is the ordinary calibration-sample error; the second term is the additional simulation error from using  $m$  bounded unbiased estimates per example-action pair. It is still a full-information statement: the analyst obtains bounded unbiased estimates for every action needed to evaluate every policy in the finite class. Bandit logging, adaptively stopped estimates, biased verifiers, or hardware timing noise that is not well modeled by bounded independent estimates require separate arguments. Here one Monte Carlo draw  $W$  should be read as an evaluation of the complete action outcome, including any internal generation, verification, and realized cost for that action. It is not a single candidate sample inside a larger best-of- $k$  action unless that single sample is itself the action.

**Corollary 6** (ERM from logged actions with known propensities). *Fix  $\lambda$  and a finite policy class  $\Pi$  independently of the logged calibration sample. Let  $z_a(x, y) = \ell_a(x, y) + \lambda c_a(x) \in [0, B]$ . Suppose the logged sample consists of iid tuples  $(X_i, Y_i, A_i, Z_i)$  such that  $(X_i, Y_i) \sim P$ , the logged action is drawn from a known behavior distribution  $\mu(\cdot \mid X_i)$ , and*

$$\mu(a \mid x) \geq \rho > 0 \quad \text{for every } a \in \mathcal{A}, x \in \mathcal{X}. \quad (36)$$

*Assume also that, conditional on  $(X_i, Y_i, A_i = a)$ , the observed logged objective satisfies  $Z_i \in [0, B]$  and*

$$\mathbb{E}[Z_i \mid X_i, Y_i, A_i = a] = z_a(X_i, Y_i). \quad (37)$$

*Define the inverse-propensity estimate*

$$\hat{J}_{\lambda, S}^{\text{ips}}(\pi) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbf{1}\{A_i = \pi(X_i)\} Z_i}{\mu(A_i \mid X_i)}. \quad (38)$$

*Let  $\hat{\pi}_{\text{ips}} \in \arg \min_{\pi \in \Pi} \hat{J}_{\lambda, S}^{\text{ips}}(\pi)$ . Then, with probability at least  $1 - \delta$  over the logged calibration sample,*

$$J_{\lambda}(\hat{\pi}_{\text{ips}}) \leq \inf_{\pi \in \Pi} J_{\lambda}(\pi) + 2B \sqrt{\frac{2 \log(2|\Pi|/\delta)}{n\rho}} + \frac{4B \log(2|\Pi|/\delta)}{3n\rho}. \quad (39)$$

*Proof.* For a fixed policy  $\pi$ , define

$$U_i(\pi) = \frac{\mathbf{1}\{A_i = \pi(X_i)\}Z_i}{\mu(A_i | X_i)}. \quad (40)$$

The support assumption gives  $U_i(\pi) \in [0, B/\rho]$ . Moreover, using the known randomized logging rule,

$$\mathbb{E}[U_i(\pi) | X_i, Y_i] = z_{\pi(X_i)}(X_i, Y_i), \quad (41)$$

and therefore  $\mathbb{E}U_i(\pi) = J_\lambda(\pi)$ . The second moment is at most  $B^2/\rho$ , because

$$\mathbb{E}\left[\frac{\mathbf{1}\{A_i = \pi(X_i)\}}{\mu(A_i | X_i)^2} \middle| X_i, Y_i\right] = \frac{1}{\mu(\pi(X_i) | X_i)} \leq \frac{1}{\rho}.$$

Bernstein's inequality and a union bound over  $\Pi$  imply that, with probability at least  $1 - \delta$ ,

$$\sup_{\pi \in \Pi} \left| \hat{J}_{\lambda, S}^{\text{ips}}(\pi) - J_\lambda(\pi) \right| \leq B \sqrt{\frac{2 \log(2|\Pi|/\delta)}{n\rho}} + \frac{2B \log(2|\Pi|/\delta)}{3n\rho}. \quad (42)$$

The ERM argument doubles this uniform deviation.  $\square$

**Proposition 2** (Logged-feedback overlap is unavoidable). *The overlap dependence in logged compute allocation is not only an artifact of the IPS analysis. Fix any  $\rho \in (0, 1/2]$  and  $\Delta \in (0, 1/4]$ . There is a one-input, two-action, zero-cost allocation problem with policy class  $\Pi = \{\pi_0, \pi_1\}$ , where  $\pi_a$  always chooses action  $a$ , and a known logging policy with  $\mu(1) = \rho$  such that the following holds. Any algorithm that observes  $n$  iid logged samples and returns a policy  $\hat{\pi}$  must, for at least one of two problem instances, suffer expected excess cost-augmented risk at least  $\Delta/4$  whenever*

$$n < \frac{1}{32\rho\Delta^2}. \quad (43)$$

*Under full-information calibration on the same two instances, the identity of the better action can be recovered with probability at least  $1 - \delta$  from  $O(\Delta^{-2} \log(1/\delta))$  samples.*

*Proof.* Let  $X = x_0$  almost surely and let the loss of action 0 be deterministically  $1/2$ . Consider two instances that differ only in the loss distribution of action 1. In instance  $+$ , action 1 has Bernoulli loss with mean  $1/2 - \Delta$ ; in instance  $-$ , it has Bernoulli loss with mean  $1/2 + \Delta$ . Thus  $\pi_1$  is better by  $\Delta$  in instance  $+$ , while  $\pi_0$  is better by  $\Delta$  in instance  $-$ .

The logging policy chooses action 1 with probability  $\rho$  and action 0 otherwise. Since action 0 has the same distribution in both instances, the KL divergence between the two one-sample logged-data distributions is

$$\rho \text{KL}(\text{Bernoulli}(1/2 - \Delta) \parallel \text{Bernoulli}(1/2 + \Delta)) \leq 16\rho\Delta^2, \quad (44)$$

where the last inequality uses  $\Delta \leq 1/4$ . Hence the KL divergence between the  $n$ -sample logged distributions is at most  $16n\rho\Delta^2$ , and Pinsker's inequality gives total variation at most  $\sqrt{8n\rho\Delta^2}$ . If  $n < 1/(32\rho\Delta^2)$ , this total variation is below  $1/2$ . Le Cam's two-point argument then implies that the sum of the two error probabilities for deciding which policy is better is at least  $1/2$ . Therefore, in at least one instance, the algorithm selects the suboptimal policy with probability at least  $1/4$ , giving expected excess risk at least  $\Delta/4$ .

For full-information calibration, every example reveals a sample from action 1 regardless of the logging probability. A Hoeffding test of whether the mean loss of action 1 is below or above  $1/2$  identifies the correct instance with  $O(\Delta^{-2} \log(1/\delta))$  samples.  $\square$

This logged-feedback result is intentionally narrow. It assumes randomized logging with known propensities, overlap for every action a candidate policy may choose, and no hidden confounding beyond the observed input used by the logging rule; equivalently, all variables used to choose actions and relevant to potential outcomes are included in  $X$ . The  $1/\rho$  factor shows the price of weak exploration, and Proposition 2 shows that this dependence is unavoidable in the simplest fixed-action allocation problem. A nested variant is handled next. Self-normalized and doubly robust estimators require separate bias and concentration arguments; the empirical terminal SNIPS diagnostics below do not change the support event. Sequential or censored logs, such as early-exit traces where running to a deeper action reveals partial information about shallower actions, require a different estimator rather than the one-action IPS estimator above.

**Definition 3** (Prefix-observing compute log). *For ordered actions  $\{0, \dots, K\}$ , a log record  $O_i = (X_i, A_i, \{Z_{i,k} : k \leq A_i\})$  is prefix-observing if each recorded  $Z_{i,k}$  is the same bounded loss-plus-cost objective that would be observed by stopping at depth  $k$ , and is invariant to later computation.*

Cached classifier checkpoints and prefix vote counts satisfy this invariance; generated outputs whose earlier answer is revised by later computation do not.

**Theorem 6** (Survival-vs-terminal separation for prefix logs). *Fix  $0 < \eta \leq s \leq 1$ ,  $\Delta \in (0, 1/4]$ , and any  $\delta \in (0, 1)$ . There is a one-input, three-action, zero-cost allocation problem with policy class  $\Pi = \{\pi_0, \pi_1\}$ , where  $\pi_a$  always chooses action  $a$  and action 2 is only a deeper logging action, and a known maximum-depth logger independent of the potential objectives such that  $P(A = 1) = \eta$  and  $P(A \geq 1) = s$ , for which the following holds.*

*First, an instance-wise lower bound for this family: any algorithm that observes  $n$  iid nested logged examples and returns a policy  $\hat{\pi}$  must, for at least one of two problem instances, suffer expected excess risk at least  $\Delta/4$  whenever*

$$n < \frac{1}{32s\Delta^2}. \quad (45)$$

*Conversely, under nested prefix feedback, which reveals  $Z_{i,1}$  whenever  $A_i \geq 1$ , a threshold test comparing the action-1 prefix mean to the known action-0 mean  $1/2$  identifies the better policy with probability at least  $1 - \delta$  from*

$$n \geq \frac{8}{s\Delta^2} \log \frac{4}{\delta} \quad (46)$$

*logged examples.*

*Second, if an algorithm first reduces the same iid nested logged examples to terminal-action feedback and observes only  $(A_i, Z_{i,A_i})$ , then for at least one of the same two instances it suffers expected excess risk at least  $\Delta/4$  whenever*

$$n < \frac{1}{32\eta\Delta^2}. \quad (47)$$

*Hence the relevant exploration mass for a prefix-observing stopping policy is survival mass  $s$ , not terminal mass  $\eta$ ; the gap can be arbitrarily large when deeper runs are common but terminal stops at the decisive prefix are rare.*

*Proof.* Let  $X = x_0$  almost surely. The logger draws a maximum reached action  $A \in \{0, 1, 2\}$  with

$$P(A = 0) = 1 - s, \quad P(A = 1) = \eta, \quad P(A = 2) = s - \eta. \quad (48)$$

Running to depth 2 reveals the prefix outcome for action 1 under nested feedback, but the terminal-only observer sees only the depth-2 outcome. Set the loss of action 0 to be deterministically  $1/2$ ,

and set  $Z_2$  deterministically equal to  $1/2$  in both instances, independent of  $Z_1$ . The two instances differ only in the potential prefix loss of action 1: in instance  $+$ ,  $Z_1 \sim \text{Bernoulli}(1/2 - \Delta)$ , while in instance  $-$ ,  $Z_1 \sim \text{Bernoulli}(1/2 + \Delta)$ . Thus  $\pi_1$  is better by  $\Delta$  in instance  $+$ , and  $\pi_0$  is better by  $\Delta$  in instance  $-$ .

For full nested feedback, the two observed one-sample distributions differ only on the event  $A \geq 1$ , whose probability is  $s$ . Therefore

$$\text{KL}(P_+^{\text{nest}} \| P_-^{\text{nest}}) = s \text{KL}(\text{Bernoulli}(1/2 - \Delta) \| \text{Bernoulli}(1/2 + \Delta)) \leq 16s\Delta^2. \quad (49)$$

Pinsker's inequality and Le Cam's two-point argument, exactly as in Proposition 2, imply the nested lower bound.

For terminal-only feedback, the two observed one-sample distributions differ only on the event  $A = 1$ , whose probability is  $\eta$ . Therefore

$$\text{KL}(P_+^{\text{term}} \| P_-^{\text{term}}) = \eta \text{KL}(\text{Bernoulli}(1/2 - \Delta) \| \text{Bernoulli}(1/2 + \Delta)) \leq 16\eta\Delta^2. \quad (50)$$

The same Le Cam argument as in Proposition 2 then gives the terminal-only lower bound.

For nested prefix feedback, the analyst observes an iid sample from action 1 on each event  $A_i \geq 1$ , which has probability  $s$ . Let  $M = \sum_{i=1}^n \mathbf{1}\{A_i \geq 1\}$ . A test that compares the empirical mean of the observed action-1 prefix losses to  $1/2$  fails with probability at most  $\exp(-ns/8) + \exp(-ns\Delta^2)$ : the first term controls  $P(M < ns/2)$  and the second applies Hoeffding's inequality conditional on  $M \geq ns/2$ . The displayed sample size makes this error at most  $\delta$ .  $\square$

Theorem 6 is the sense in which nested compute logs are not merely ordinary bandit logs with different notation. A run that continues beyond a candidate stopping action can still identify the objective of that stopping action, so discarding prefix outcomes can lose a factor  $s/\eta$  in information.

**Corollary 7** (ERM from nested sequential compute logs). *Let the compute actions be ordered as  $\mathcal{A} = \{0, 1, \dots, K\}$  and let  $\Pi$  be a finite class of stopping policies  $\pi : \mathcal{X} \rightarrow \{0, \dots, K\}$  fixed independently of the logged calibration sample. Suppose the logged calibration sample consists of iid draws. For each example there are potential prefix objectives  $Z_{i,0}, \dots, Z_{i,K} \in [0, B]$  satisfying*

$$\mathbb{E}[Z_{i,k} \mid X_i = x, Y_i = y] = z_k(x, y) = \ell_k(x, y) + \lambda c_k(x). \quad (51)$$

*The log records a maximum reached action  $A_i$  and reveals  $Z_{i,k}$  if and only if  $A_i \geq k$ . Let the known survival propensity be*

$$s_k(x) = P(A_i \geq k \mid X_i = x), \quad (52)$$

*and assume policy overlap*

$$\rho = \inf_{\pi \in \Pi} \text{ess inf}_X s_{\pi(X)}(X) > 0. \quad (53)$$

*Finally, assume prefix ignorability, or coarsening at random:*

$$\mathbb{E} \left[ \frac{\mathbf{1}\{A_i \geq k\} Z_{i,k}}{s_k(X_i)} \mid X_i, Y_i \right] = z_k(X_i, Y_i) \quad \text{for every } k. \quad (54)$$

*A sufficient primitive condition is that target policies are fixed before calibration and  $X$ -measurable, survival propensities are positive wherever such policies may stop, and conditional on  $X_i$  the event  $\{A_i \geq k\}$  is independent of  $(Y_i, Z_{i,0}, \dots, Z_{i,K})$  with probability  $s_k(X_i)$ . If continuation depends on recorded intermediate outputs, the conditioning variable must be the observed reachable history and*

the corresponding sequential coarsening condition replaces this marginal survival condition. Define the nested inverse-survival estimate

$$\hat{J}_{\lambda,S}^{\text{nest}}(\pi) = \frac{1}{n} \sum_{i=1}^n \frac{\mathbf{1}\{A_i \geq \pi(X_i)\} Z_{i,\pi(X_i)}}{s_{\pi(X_i)}(X_i)}. \quad (55)$$

If  $\hat{\pi}_{\text{nest}} \in \arg \min_{\pi \in \Pi} \hat{J}_{\lambda,S}^{\text{nest}}(\pi)$ , then, with probability at least  $1 - \delta$  over the logged sample,

$$J_{\lambda}(\hat{\pi}_{\text{nest}}) \leq \inf_{\pi \in \Pi} J_{\lambda}(\pi) + 2B \sqrt{\frac{2 \log(2|\Pi|/\delta)}{n\rho}} + \frac{4B \log(2|\Pi|/\delta)}{3n\rho}. \quad (56)$$

*Proof.* For a fixed stopping policy  $\pi$ , define

$$V_i(\pi) = \frac{\mathbf{1}\{A_i \geq \pi(X_i)\} Z_{i,\pi(X_i)}}{s_{\pi(X_i)}(X_i)}. \quad (57)$$

By policy overlap,  $V_i(\pi) \in [0, B/\rho]$ . By prefix ignorability,

$$\mathbb{E}[V_i(\pi) \mid X_i, Y_i] = z_{\pi(X_i)}(X_i, Y_i), \quad (58)$$

so  $\mathbb{E}V_i(\pi) = J_{\lambda}(\pi)$ . Also

$$\mathbb{E}[V_i(\pi)^2 \mid X_i, Y_i] \leq \frac{B^2}{s_{\pi(X_i)}(X_i)} \leq \frac{B^2}{\rho}. \quad (59)$$

Bernstein's inequality and a union bound over  $\Pi$  give, with probability at least  $1 - \delta$ ,

$$\sup_{\pi \in \Pi} \left| \hat{J}_{\lambda,S}^{\text{nest}}(\pi) - J_{\lambda}(\pi) \right| \leq B \sqrt{\frac{2 \log(2|\Pi|/\delta)}{n\rho}} + \frac{2B \log(2|\Pi|/\delta)}{3n\rho}. \quad (60)$$

The usual ERM argument doubles this uniform deviation.  $\square$

The denominator in the nested estimator is a survival probability, not a terminal-action probability. If  $\mu(a \mid x) = P(A_i = a \mid X_i = x)$  denotes the distribution of the maximum reached action, ordinary one-action feedback uses  $\mu(k \mid x)$ , while nested prefix feedback uses

$$s_k(x) = P(A_i \geq k \mid X_i = x) = \sum_{a=k}^K \mu(a \mid x). \quad (61)$$

Thus one run to depth  $a$  supplies usable observations for every prefix  $k \leq a$ . The assumption that makes this valid is prefix ignorability: if continuation depends on intermediate losses, verifier scores, or confidences not included in  $X$ , then marginal survival propensities  $s_k(x)$  are not enough and history-dependent sequential propensities are needed.

**Proposition 3** (Sequential continuation propensities). *The nested ERM bound extends to contextual or history-dependent continuation rules when the logged continuation propensities are known. Let  $C_{i,k}$  be the indicator that example  $i$  survives to depth  $k$ , with  $C_{i,0} = 1$ , and let  $H_{i,k-1}$  be the observed history available immediately before deciding whether to continue from depth  $k-1$  to depth  $k$ . Suppose*

$$g_k(h) = P(C_{i,k} = 1 \mid C_{i,k-1} = 1, H_{i,k-1} = h) \quad (62)$$

is known, and define

$$G_{i,k} = \prod_{j=1}^k g_j(H_{i,j-1}), \quad G_{i,0} = 1. \quad (63)$$

Assume  $Z_{i,k} \in [0, B]$  is observed when  $C_{i,k} = 1$ , the policy class  $\Pi$  consists of  $X$ -measurable stopping policies  $\pi : \mathcal{X} \rightarrow \{0, \dots, K\}$  fixed before calibration, and sequential coarsening at random holds:

$$\mathbb{E} \left[ \frac{C_{i,k} Z_{i,k}}{G_{i,k}} \middle| X_i, Y_i \right] = z_k(X_i, Y_i) \quad \text{for every } k. \quad (64)$$

One primitive sufficient condition is that all variables used by the logger are included in  $H$ , and for every target depth  $k$  and continuation step  $j \leq k$ , on reachable histories,

$$C_{i,j} \perp\!\!\!\perp Z_{i,k} \mid C_{i,j-1} = 1, H_{i,j-1}, \quad g_j(H_{i,j-1}) > 0. \quad (65)$$

If

$$\rho = \inf_{\pi \in \Pi} \text{ess inf}_{(X,H)} G_{i,\pi(X_i)} > 0 \quad (66)$$

and  $\hat{\pi}_{\text{seq}}$  minimizes

$$\hat{J}_{\lambda, S}^{\text{seq}}(\pi) = \frac{1}{n} \sum_{i=1}^n \frac{C_{i,\pi(X_i)} Z_{i,\pi(X_i)}}{G_{i,\pi(X_i)}}, \quad (67)$$

then, with probability at least  $1 - \delta$ ,

$$J_{\lambda}(\hat{\pi}_{\text{seq}}) \leq \inf_{\pi \in \Pi} J_{\lambda}(\pi) + 2B \sqrt{\frac{2 \log(2|\Pi|/\delta)}{n\rho}} + \frac{4B \log(2|\Pi|/\delta)}{3n\rho}. \quad (68)$$

*Proof.* For each fixed  $\pi$ , the summand  $C_{i,\pi(X_i)} Z_{i,\pi(X_i)} / G_{i,\pi(X_i)}$  lies in  $[0, B/\rho]$  and has expectation  $J_{\lambda}(\pi)$  by sequential coarsening at random. Its second moment is at most  $B^2/\rho$ . Bernstein's inequality and a union bound over  $\Pi$  give a uniform deviation bound with radius

$$B \sqrt{\frac{2 \log(2|\Pi|/\delta)}{n\rho}} + \frac{2B \log(2|\Pi|/\delta)}{3n\rho}, \quad (69)$$

and the standard ERM comparison doubles it.  $\square$

Target policies in Proposition 3 choose a depth from  $X$  before execution. History may affect the logger's continuation rule; the bound does not cover deployment policies that adapt after seeing intermediate states.

Using a marginal survival probability in a contextual logger is generally biased. For a one-depth example, let  $X \in \{0, 1\}$  be uniform, let  $Z_1 = \mathbf{1}\{X = 1\}$ , and let  $C_1$  be drawn with  $P(C_1 = 1 \mid X = 1) = 1$  and  $P(C_1 = 1 \mid X = 0) = \alpha$  for some  $0 < \alpha < 1$ . The target is  $\mathbb{E}Z_1 = 1/2$ . The contextual estimator  $C_1 Z_1 / P(C_1 = 1 \mid X)$  is unbiased, but the marginal inverse-weighted estimator using  $\bar{s} = P(C_1 = 1) = (1 + \alpha)/2$  has expectation  $1/(1 + \alpha)$ , which differs from  $1/2$  unless the logger is context independent. This is the formal version of the marginal-survival ablation in the contextual text-routing experiment.

**Corollary 8** (Clipped IPS ERM with explicit bias). *Fix  $\lambda$  and a finite policy class  $\Pi$  independently of the logged calibration sample. Let  $z_a(x, y) = \ell_a(x, y) + \lambda c_a(x) \in [0, B]$ . Suppose the logged sample consists of iid tuples  $(X_i, Y_i, A_i, Z_i)$  such that  $(X_i, Y_i) \sim P$ , the logged action is drawn from a*

known behavior distribution  $\mu(\cdot \mid X_i)$ , and conditional on  $(X_i, Y_i, A_i = a)$  the observed logged objective satisfies  $Z_i \in [0, B]$  and

$$\mathbb{E}[Z_i \mid X_i, Y_i, A_i = a] = z_a(X_i, Y_i). \quad (70)$$

Assume policy support,  $\mu(\pi(x) \mid x) > 0$  for every  $\pi \in \Pi$  and  $P_X$ -almost every  $x$ , but do not require a uniform lower bound on this propensity. For a clip level  $\tau > 0$ , define

$$\hat{J}_{\lambda, S, \tau}^{\text{clip}}(\pi) = \frac{1}{n} \sum_{i=1}^n \min \left\{ \frac{\mathbf{1}\{A_i = \pi(X_i)\}}{\mu(A_i \mid X_i)}, \tau \right\} Z_i. \quad (71)$$

Let

$$J_{\lambda, \tau}^{\text{clip}}(\pi) = \mathbb{E} [z_{\pi(X)}(X, Y) \min\{1, \tau \mu(\pi(X) \mid X)\}], \quad (72)$$

and define the clipping bias

$$b_\tau(\pi) = J_\lambda(\pi) - J_{\lambda, \tau}^{\text{clip}}(\pi) = \mathbb{E} [z_{\pi(X)}(X, Y)(1 - \tau \mu(\pi(X) \mid X))_+]. \quad (73)$$

If  $\hat{\pi}_\tau \in \arg \min_{\pi \in \Pi} \hat{J}_{\lambda, S, \tau}^{\text{clip}}(\pi)$ , then with probability at least  $1 - \delta$ ,

$$J_{\lambda, \tau}^{\text{clip}}(\hat{\pi}_\tau) \leq \inf_{\pi \in \Pi} J_{\lambda, \tau}^{\text{clip}}(\pi) + 2B\tau \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (74)$$

Equivalently,

$$J_\lambda(\hat{\pi}_\tau) \leq \inf_{\pi \in \Pi} \{J_\lambda(\pi) - b_\tau(\pi)\} + b_\tau(\hat{\pi}_\tau) + 2B\tau \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (75)$$

In particular,

$$J_\lambda(\hat{\pi}_\tau) \leq \inf_{\pi \in \Pi} J_\lambda(\pi) + b_\tau(\hat{\pi}_\tau) + 2B\tau \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (76)$$

If  $\beta_\tau = \sup_{\pi \in \Pi} b_\tau(\pi)$ , then

$$J_\lambda(\hat{\pi}_\tau) \leq \inf_{\pi \in \Pi} J_\lambda(\pi) + \beta_\tau + 2B\tau \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (77)$$

*Proof.* For a fixed  $\pi$ , the clipped summand is in  $[0, B\tau]$ . Its expectation is  $J_{\lambda, \tau}^{\text{clip}}(\pi)$ , because conditional on  $(X, Y)$  only the event  $A = \pi(X)$  contributes:

$$\mathbb{E} \left[ \min \left\{ \frac{\mathbf{1}\{A = \pi(X)\}}{\mu(A \mid X)}, \tau \right\} Z \mid X, Y \right] = z_{\pi(X)}(X, Y) \min\{1, \tau \mu(\pi(X) \mid X)\}. \quad (78)$$

Hoeffding's inequality and a union bound over  $\Pi$  imply that, with probability at least  $1 - \delta$ ,

$$\sup_{\pi \in \Pi} \left| \hat{J}_{\lambda, S, \tau}^{\text{clip}}(\pi) - J_{\lambda, \tau}^{\text{clip}}(\pi) \right| \leq B\tau \sqrt{\frac{\log(2|\Pi|/\delta)}{2n}}. \quad (79)$$

On this event, the ERM argument gives, for every  $\pi \in \Pi$ ,

$$J_{\lambda, \tau}^{\text{clip}}(\hat{\pi}_\tau) \leq J_{\lambda, \tau}^{\text{clip}}(\pi) + 2\epsilon_\tau, \quad (80)$$

where  $\epsilon_\tau$  is the displayed uniform deviation. Taking the infimum over  $\pi$  gives the clipped-objective oracle inequality. Since  $J_\lambda(\pi) = J_{\lambda, \tau}^{\text{clip}}(\pi) + b_\tau(\pi)$ , the two true-risk forms follow by rewriting this inequality and then using  $J_{\lambda, \tau}^{\text{clip}}(\pi) \leq J_\lambda(\pi)$ .  $\square$



Table 2: Finite-class calibration guarantees in this paper. Each row applies to a fixed finite compute-policy class and bounded cost-augmented objectives.

| Feedback                     | Estimator                | Statistical price                                | Main caveat                                         |
|------------------------------|--------------------------|--------------------------------------------------|-----------------------------------------------------|
| Full information             | empirical mean           | $B\sqrt{\log  \Pi /n}$                           | evaluates every action                              |
| Monte Carlo full information | bounded action estimates | $B\sqrt{\log  \Pi /n} + B\sqrt{\log  \Pi /(nm)}$ | all actions estimated independently                 |
| Logged actions               | IPS                      | $B\sqrt{\log  \Pi /(n\rho)}$                     | known propensities and overlap                      |
| Nested sequential logs       | inverse-survival IPS     | $B\sqrt{\log  \Pi /(n\rho)}$                     | known survival propensities and prefix ignorability |
| Logged actions               | clipped IPS              | $B\tau\sqrt{\log  \Pi /n} + \beta_\tau$          | biased target objective                             |

Clipping trades variance for downward bias. If  $\tau \geq 1/\rho$  under the overlap condition of Corollary 6, then  $b_\tau(\pi) = 0$  for all  $\pi$  and the clipped estimator reduces to ordinary IPS. Smaller  $\tau$  can substantially reduce empirical variance but changes the target objective; the selected policy must then be judged with the bias term included.

These finite-class variants are not meant to be surprising. Their role is to make explicit that adaptive inference introduces a policy class whose complexity and feedback model matter, even if the underlying predictors are fixed.

Theorem 3 assumes binwise action-objective estimates and then uses an argmin-stability argument. A related but distinct guarantee treats the entire bin-to-action assignment as a member of a finite policy class and performs empirical-risk selection over that class. This gives a finite-sample guarantee for a common calibration procedure: fix a difficulty partition, then use calibration data to choose an allowed bin-to-action assignment.

**Corollary 9** (Finite-sample binned ERM). *Let  $b : \mathcal{X} \rightarrow \{1, \dots, M\}$  be a measurable binning chosen independently of the calibration sample  $S$ . Let  $\Pi_b$  be the set of all policies that are constant on each bin, so  $|\Pi_b| \leq |\mathcal{A}|^M$ . Let*

$$\hat{\pi}_b \in \arg \min_{\pi \in \Pi_b} \hat{J}_{\lambda, S}(\pi), \quad \pi_b^* \in \arg \min_{\pi \in \Pi_b} J_\lambda(\pi). \quad (81)$$

Then, with probability at least  $1 - \delta$  over  $S \sim P^n$ ,

$$J_\lambda(\hat{\pi}_b) - J_\lambda(\pi_b^*) \leq 2B\sqrt{\frac{M \log |\mathcal{A}| + \log(2/\delta)}{2n}}. \quad (82)$$

Consequently, relative to any unrestricted oracle policy  $\pi^*$  that minimizes  $J_\lambda$  over all measurable compute policies,

$$J_\lambda(\hat{\pi}_b) - J_\lambda(\pi^*) \leq \underbrace{J_\lambda(\pi_b^*) - J_\lambda(\pi^*)}_{\text{binning approximation error}} + 2B\sqrt{\frac{M \log |\mathcal{A}| + \log(2/\delta)}{2n}}. \quad (83)$$

If the search is restricted to a subset  $\Gamma \subseteq \mathcal{A}^M$  of allowed bin-to-action assignments, such as monotone assignments, the same statement holds with  $M \log |\mathcal{A}|$  replaced by  $\log |\Gamma|$  and with  $\pi_b^*$  defined as the best policy in that restricted class.

*Proof.* Apply Theorem 5 to the finite class  $\Pi_b$ . Since  $|\Pi_b| \leq |\mathcal{A}|^M$ , the event

$$\sup_{\pi \in \Pi_b} |J_\lambda(\pi) - \hat{J}_{\lambda, S}(\pi)| \leq B\sqrt{\frac{M \log |\mathcal{A}| + \log(2/\delta)}{2n}} \quad (84)$$

has probability at least  $1 - \delta$ . On this event, empirical optimality gives the standard ERM inequality  $J_\lambda(\hat{\pi}_b) - J_\lambda(\pi_b^*) \leq 2 \sup_{\pi \in \Pi_b} |J_\lambda(\pi) - \hat{J}_{\lambda, S}(\pi)|$ . Adding and subtracting  $J_\lambda(\pi_b^*)$  gives the oracle decomposition. The restricted-class variant is identical with  $|\Pi_b|$  replaced by  $|\Gamma|$ .  $\square$

Corollary 9 is not a new learning-theoretic tool; it is the direct finite-sample counterpart of the plug-in calibration view used later in the experiments. It does not say that the chosen bins are optimal; that difficulty-estimation problem is isolated in the approximation term. It says that, once the bins, action family, and costs are fixed on data independent of the calibration split, the calibration split can select among the resulting compute policies with an explicit sample-complexity term. If bins, difficulty scores, or cost normalizers are learned from the same calibration sample, one needs either an independent construction split or a uniform-convergence argument over that additional choice class.

For a finite set of candidate binnings, the needed uniform-convergence argument is still elementary.

**Corollary 10** (Finite learned-binning class). *Let  $\mathcal{B}$  be a nonempty finite collection of measurable maps  $b : \mathcal{X} \rightarrow \{1, \dots, M\}$ , chosen independently of the calibration sample. Define*

$$\Pi_{\mathcal{B}} = \{\pi : \pi(x) = g(b(x)), \ b \in \mathcal{B}, \ g : \{1, \dots, M\} \rightarrow \mathcal{A}\}. \quad (85)$$

*Let  $\hat{\pi}_{\mathcal{B}} \in \arg \min_{\pi \in \Pi_{\mathcal{B}}} \hat{J}_{\lambda, S}(\pi)$  and let  $\pi_{\mathcal{B}}^* \in \arg \min_{\pi \in \Pi_{\mathcal{B}}} J_{\lambda}(\pi)$ . Using the same range constant  $B = 1 + \lambda C$  as in Theorem 5, with probability at least  $1 - \delta$  over the calibration sample,*

$$J_{\lambda}(\hat{\pi}_{\mathcal{B}}) - J_{\lambda}(\pi_{\mathcal{B}}^*) \leq 2B \sqrt{\frac{\log |\mathcal{B}| + M \log |\mathcal{A}| + \log(2/\delta)}{2n}}. \quad (86)$$

*If each binning is paired with a restricted assignment set  $\Gamma_b \subseteq \mathcal{A}^M$ , the term  $\log |\mathcal{B}| + M \log |\mathcal{A}|$  may be replaced by  $\log(\sum_{b \in \mathcal{B}} |\Gamma_b|)$ .*

*Proof.* The policy class  $\Pi_{\mathcal{B}}$  is finite and has cardinality at most  $|\mathcal{B}| |\mathcal{A}|^M$ . Applying the ERM corollary after Theorem 5 to this class gives the displayed bound. The restricted-assignment version counts the smaller class  $\{(b, g) : b \in \mathcal{B}, \ g \in \Gamma_b\}$ .  $\square$

Corollary 10 covers only finite candidate collections. It is useful for model-selection style calibration, such as choosing among a finite grid of difficulty scores, thresholds, or monotonicity constraints. It does not cover an arbitrary continuous family of learned scores, nor a candidate set generated from the same calibration sample, without replacing the cardinality term by an appropriate capacity bound or using a separate construction split. This extension is included to clarify what would be needed if calibration were used to choose among several candidate difficulty partitions; it is not used in the digits experiments, where the binning rule is fixed from the training split before calibration.

## 6 Average Compute Budgets

Many deployments specify an average compute budget rather than a fixed exchange-rate parameter  $\lambda$ . Let

$$R(\pi) = \mathbb{E}[\ell_{\pi(X)}(X, Y)], \quad C(\pi) = \mathbb{E}[c_{\pi(X)}(X)]. \quad (87)$$

The constrained problem is

$$\min_{\pi \in \Pi} R(\pi) \quad \text{subject to} \quad C(\pi) \leq b. \quad (88)$$

The cost-augmented objective is the Lagrangian relaxation  $R(\pi) + \lambda(C(\pi) - b)$  up to the constant  $-\lambda b$ .

**Proposition 4** (Lagrangian link to budgeted prediction). *Let  $\pi_\lambda \in \arg \min_{\pi \in \Pi} R(\pi) + \lambda C(\pi)$  for some  $\lambda \geq 0$ . For any policy  $\pi$  satisfying  $C(\pi) \leq b$ ,*

$$R(\pi_\lambda) - R(\pi) \leq \lambda (C(\pi) - C(\pi_\lambda)) \leq \lambda (b - C(\pi_\lambda)). \quad (89)$$

*In particular, if  $C(\pi_\lambda) = b$ , then  $\pi_\lambda$  is optimal for the budget-constrained problem over  $\Pi$ .*

*Proof.* The optimality of  $\pi_\lambda$  gives  $R(\pi_\lambda) + \lambda C(\pi_\lambda) \leq R(\pi) + \lambda C(\pi)$  for every  $\pi \in \Pi$ . Rearranging proves the first inequality; feasibility of  $\pi$  gives the second. If  $C(\pi_\lambda) = b$ , the right-hand side is nonpositive for every feasible  $\pi$ , so no feasible policy has smaller risk.  $\square$

In a finite or otherwise nonconvex policy class, a sweep over  $\lambda$  need not realize every possible budget. It identifies supported accuracy-cost tradeoff points. Exact budget matching may require randomizing between policies or otherwise convexifying the policy class; the deterministic statement above only gives exact constrained optimality when the selected policy itself attains the target budget.

## 7 Ordered Computation and Marginal Utility

Many adaptive systems have ordered compute actions, such as exit depth  $0, 1, \dots, K$  or number of samples. In this case, the oracle policy has a simple marginal-utility interpretation.

**Assumption 1** (Ordered diminishing returns). *Let  $\mathcal{A} = \{0, 1, \dots, K\}$ . For each  $x$ , assume costs are increasing and risks are nonincreasing:  $c_{k+1}(x) > c_k(x)$  and  $r_{k+1}(x) \leq r_k(x)$ . Define marginal improvement  $\Delta_k(x) = r_k(x) - r_{k+1}(x)$  and marginal cost  $\Gamma_k(x) = c_{k+1}(x) - c_k(x)$ . Assume the ratios  $\Delta_k(x)/\Gamma_k(x)$  are nonincreasing in  $k$ .*

**Proposition 5** (Marginal-utility stopping). *Under the ordered diminishing returns assumption, define  $k_{<}(x)$  as the first index  $k$  for which*

$$\Delta_k(x) \leq \lambda \Gamma_k(x), \quad (90)$$

*or set  $k_{<}(x) = K$  if no such index exists. Then  $k_{<}(x)$  is the cheapest oracle minimizer of  $r_k(x) + \lambda c_k(x)$ . Computation continues exactly while the expected loss reduction strictly exceeds the cost charged by  $\lambda$ ; equality creates multiple oracle minimizers.*

*Proof.* Moving from  $k$  to  $k+1$  changes the cost-augmented objective by  $-\Delta_k(x) + \lambda \Gamma_k(x)$ . The move is beneficial iff this quantity is negative. Since  $\Delta_k(x)/\Gamma_k(x)$  is nonincreasing, once the inequality  $\Delta_k(x) > \lambda \Gamma_k(x)$  fails, no later step can be strictly beneficial. Thus the first failure point is a global minimizer; using the first such index chooses the cheapest minimizer when ties occur.  $\square$

This proposition links the formal objective to a common engineering rule: spend one more unit of compute only when its predicted marginal benefit exceeds its price. Theorem 1 then quantifies the penalty for estimating that marginal benefit imperfectly.

## 8 Verifier-Guided Resampling

Repeated sampling with a verifier is a common test-time scaling mechanism. The compute action is the number of samples to draw, but the value of additional samples depends on verifier quality. A simple model already shows why “sample more” is not an unconditional principle.

Assume each candidate is independently correct with probability  $p$ . A binary verifier accepts a correct candidate with probability  $t$  and accepts an incorrect candidate with probability  $f$ . Let

$$\alpha = pt + (1 - p)f, \quad \beta = pt, \quad r = 1 - \alpha. \quad (91)$$

The policy samples candidates until either the verifier accepts one or  $k$  attempts have been used. If no candidate is accepted, it falls back to a baseline answer that is correct with probability  $b$ .

**Proposition 6** (False-positive ceiling and finite sampling). *If  $\alpha > 0$ , the probability that an accepted candidate is correct is*

$$\rho_{\text{ver}} = \frac{\beta}{\alpha} = \frac{pt}{pt + (1 - p)f}. \quad (92)$$

The overall correctness after at most  $k$  attempts is

$$P_k = \rho_{\text{ver}} (1 - r^k) + br^k. \quad (93)$$

The expected number of attempted samples under early stopping is

$$\mathbb{E}[N_k] = \frac{1 - r^k}{\alpha}. \quad (94)$$

Consequently,  $P_k \rightarrow \rho_{\text{ver}}$  as  $k \rightarrow \infty$ . If  $0 < p < 1$  and  $f > 0$ , then  $\rho_{\text{ver}} < 1$ , so false positives impose a ceiling that cannot be removed by more sampling under this policy. Additional samples improve raw correctness exactly when  $\rho_{\text{ver}} > b$ , hurt raw correctness exactly when  $\rho_{\text{ver}} < b$ , and are neutral when  $\rho_{\text{ver}} = b$ . With fixed cap cost  $\lambda k$ , the utility  $U_k = P_k - \lambda k$  satisfies

$$U_{k+1} - U_k = (\beta - b\alpha)r^k - \lambda. \quad (95)$$

Thus if  $\lambda > 0$ , every utility-maximizing  $k$  is finite whenever  $\rho_{\text{ver}} > b$ . With realized expected stopping cost,  $\tilde{U}_k = P_k - \lambda \mathbb{E}[N_k]$  satisfies

$$\tilde{U}_{k+1} - \tilde{U}_k = (\beta - b\alpha - \lambda)r^k. \quad (96)$$

*Proof.* Each attempt has three relevant outcomes: accepted and correct with probability  $\beta$ , accepted and incorrect with probability  $\alpha - \beta$ , and rejected with probability  $r$ . First acceptance at time  $j$  and correctness has probability  $r^{j-1}\beta$ , so summing the geometric series gives the accepted-correct contribution  $(\beta/\alpha)(1 - r^k)$ . The no-accept event has probability  $r^k$  and contributes baseline correctness  $b$ . The expected stopping cost follows from the tail-sum identity  $\mathbb{E}[N_k] = \sum_{j=1}^k P(N_k \geq j) = \sum_{j=1}^k r^{j-1}$ . The marginal formulas follow by subtraction.  $\square$

The preceding proposition assumes identical independent candidate quality. A different failure mode appears when the per-attempt success probability varies by input. This induces positive dependence among attempts in the population even if attempts are conditionally independent for a fixed input.

**Proposition 7** (Latent difficulty slows best-of- $k$ ). *Let  $\theta = \theta(X) \in [0, 1]$  be a latent per-attempt success probability. Conditional on  $X$ , let  $Z_1, Z_2, \dots$  be independent Bernoulli( $\theta$ ) indicators that an attempt produces a useful successful candidate. Define*

$$P_k = P\left(\max_{1 \leq i \leq k} Z_i = 1\right), \quad P_0 = 0, \quad (97)$$

and let  $\mu = \mathbb{E}[\theta]$ . Then

$$P_k = 1 - \mathbb{E} \left[ (1 - \theta)^k \right]. \quad (98)$$

For every  $k \geq 1$ ,

$$P_k \leq 1 - (1 - \mu)^k, \quad (99)$$

with equality for  $k = 1$ , and for  $k > 1$  equality only when  $\theta$  is almost surely constant. Moreover,

$$P_k \rightarrow P(\theta > 0) \quad \text{as } k \rightarrow \infty, \quad (100)$$

and for distinct attempts  $i \neq j$ ,

$$\text{Cov}(Z_i, Z_j) = \text{Var}(\theta). \quad (101)$$

The marginal gain from the  $(k + 1)$ st attempt is

$$P_{k+1} - P_k = \mathbb{E} \left[ \theta(1 - \theta)^k \right], \quad (102)$$

which decreases to zero. Hence, under fixed cap cost  $U_k = P_k - \lambda k$  with  $\lambda > 0$ , the utility has a finite maximizer.

*Proof.* Conditioning on  $X$ , the probability that all  $k$  attempts fail is  $(1 - \theta(X))^k$ , giving the expression for  $P_k$ . The map  $s \mapsto (1 - s)^k$  is convex on  $[0, 1]$  for  $k \geq 1$  and strictly convex for  $k > 1$ , so Jensen's inequality gives  $\mathbb{E}[(1 - \theta)^k] \geq (1 - \mathbb{E}[\theta])^k$ , with the stated equality condition. Since  $(1 - \theta)^k \rightarrow \mathbf{1}\{\theta = 0\}$  and the sequence is bounded by one, dominated convergence gives  $P_k \rightarrow 1 - P(\theta = 0) = P(\theta > 0)$ . The covariance identity follows from the law of total covariance:

$$\text{Cov}(Z_i, Z_j) = \mathbb{E}[\text{Cov}(Z_i, Z_j \mid X)] + \text{Cov}(\mathbb{E}[Z_i \mid X], \mathbb{E}[Z_j \mid X]) = \text{Var}(\theta). \quad (103)$$

The marginal formula follows by subtracting  $P_k$  from  $P_{k+1}$ . It decreases pointwise in  $k$  and converges to zero by dominated convergence. Therefore  $U_{k+1} - U_k < 0$  for all sufficiently large  $k$  whenever  $\lambda > 0$ , so a finite maximizer exists.  $\square$

These resampling results are deliberately stylized. Proposition 6 captures false-positive early stopping, while Proposition 7 captures a separate latent-difficulty mechanism; neither is a full model of graded verifier scores or adaptive search. Their role is to show that stochastic generation, verifier error, and input heterogeneity can be expressed in the same action-value language as the previous sections: the action objective for sample count  $k$  is the expected loss plus its inference cost, and the useful marginal value of more samples can vanish or become negative. If  $\alpha = 0$ , the verifier never accepts and the fallback is always used.

## 9 PAC-Bayes View

For randomized compute policies, a PAC-Bayes statement follows by applying a standard bound to the normalized cost-augmented loss. Let  $P_0$  be a sample-independent prior over deterministic policies and let  $Q \ll P_0$  be a posterior distribution over policies selected after seeing the sample. This is a Gibbs distribution that samples a policy, not a claim about all possible per-input stochastic kernels. Define

$$L(Q) = \mathbb{E}_{\pi \sim Q} [J_\lambda(\pi)/B], \quad \hat{L}_S(Q) = \mathbb{E}_{\pi \sim Q} \left[ \hat{J}_{\lambda, S}(\pi)/B \right]. \quad (104)$$

**Theorem 7** (PAC-Bayes bound for compute policies). *Let  $P_0$  be any sample-independent prior over  $\Pi$ . With probability at least  $1 - \delta$  over  $S \sim P^n$ , every posterior distribution  $Q$  over  $\Pi$  satisfying  $Q \ll P_0$  obeys*

$$L(Q) \leq \widehat{L}_S(Q) + \sqrt{\frac{\text{KL}(Q\|P_0) + \log(2\sqrt{n}/\delta)}{2n}}. \quad (105)$$

*Equivalently, multiplying by  $B$  gives a bound on the unnormalized cost-augmented risk.*

*Proof.* Apply the standard McAllester PAC-Bayes bound to the bounded loss  $(\ell_{\pi(X)}(X, Y) + \lambda c_{\pi(X)}(X))/B$ .  $\square$

The prior can encode a preference for simpler or cheaper compute policies. This does not make the PAC-Bayes mechanism new; it only clarifies how inference cost can enter the loss being bounded and the prior over allocation policies.

## 10 Empirical Checks of the Allocation Theory

The experiments below are controlled sanity checks for the preceding guarantees, not benchmark claims. Each check isolates one term in the allocation theory while keeping the action class small enough to run on a laptop. The synthetic allocation and Monte Carlo experiments test full-information objective estimation. The logged-bandit, nested-log, terminal-propensity separation, and clipped-IPS experiments remove full-information feedback and expose overlap, prefix observability, survival-versus-terminal support, and clipping bias. The resampling checks visualize verifier false positives and latent input difficulty. The text routing, sampled-answer trace, staged-boosting trace, and digits case studies instantiate finite-action calibration on small real datasets with measured or proxy CPU costs. None of these experiments is a large-scale LLM routing benchmark.

### 10.1 Synthetic Allocation Check

The theory can be probed without training a large model. Consider inputs with a latent difficulty  $d \in [0, 1]$  and ordered compute actions  $k = 0, \dots, K$ . Define a synthetic risk curve such as

$$r_k(d) = \sigma(\alpha(d - \beta_k)), \quad (106)$$

where larger  $k$  lowers the risk more strongly for difficult inputs, and let  $c_k = k/K$ . Add bounded noise to produce  $\widehat{r}_k(d)$ . Theorem 1 predicts that the excess cost-augmented risk of the estimated policy scales at most linearly with the uniform risk-curve error.

We ran this check with 20,000 synthetic inputs, six actions, and  $\lambda = 0.18$ . The repository includes the script, CSV output, and the plot in Figure 1. The observed regret grows with the risk-estimation error but remains well below the worst-case  $2\epsilon$  envelope; the mismatch rate also increases, consistent with the margin-localized view that noisy objectives primarily change decisions near action-tie boundaries.

### 10.2 Monte Carlo Action-Evaluation ERM Check

We next instantiate Corollary 5 directly. The policy class is fixed before calibration: monotone maps from five equal-width difficulty bins to four ordered actions, giving 56 candidate policies. For each synthetic input with difficulty  $d$ , action  $a$  has loss probability  $r_a(d)$ , deterministic normalized

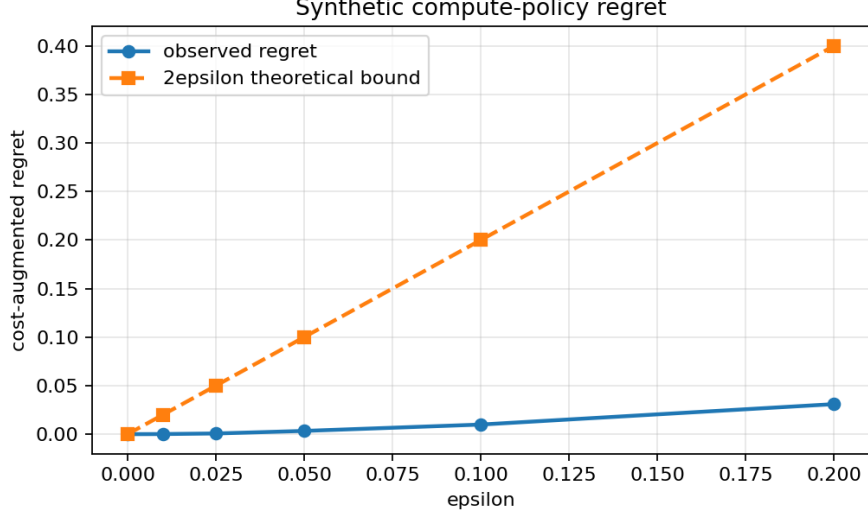


Figure 1: Synthetic regret of the estimated compute policy versus the  $2\epsilon$  theoretical envelope. The experiment is illustrative only; it is not evidence about large language models.

cost  $c_a$ , and objective  $z_a(d) = r_a(d) + \lambda c_a$  with  $\lambda = 0.18$  and range constant  $B = 1 + \lambda$ . For every calibration example, every action, and each Monte Carlo repetition, we draw

$$L_{i,a,j} \sim \text{Bernoulli}(r_a(d_i)), \quad W_{i,a,j} = L_{i,a,j} + \lambda c_a. \quad (107)$$

Thus  $W_{i,a,j} \in [0, B]$  and  $\mathbb{E}[W_{i,a,j} \mid d_i] = z_a(d_i)$ . ERM uses all action estimates  $m^{-1} \sum_{j=1}^m W_{i,a,j}$  on  $n = 120$  calibration examples, and the selected policy is evaluated on a deterministic population grid of 200,000 points. We repeat the experiment 400 times for each  $m$ .

Figure 2 shows the behavior predicted by the corollary. The population regret of the noisy ERM policy decreases from  $4.55 \times 10^{-3}$  at  $m = 1$  to zero at  $m = 64$  in this synthetic setting, while the selected-policy mismatch rate relative to full-information ERM falls from 0.318 at  $m = 1$  to zero. The proof-level quantity  $\sup_{\pi \in \Pi} |\tilde{J}(\pi) - \hat{J}(\pi)|$  also shrinks with the expected  $1/\sqrt{m}$  shape, from 0.0680 at  $m = 1$  to 0.0061 at  $m = 128$ . The finite-class Hoeffding bound covers all repetitions in this run, but is intentionally conservative: its total value is 0.883 at  $m = 1$  and 0.481 at  $m = 128$ . The full-information ERM baseline selects the same population-best policy in this configured example, so the figure isolates Monte Carlo action-evaluation noise rather than sample-selection error.

### 10.3 Logged-Bandit IPS ERM Check

The previous experiment gives the analyst noisy estimates for every action. We also ran a logged-feedback experiment matching Corollary 6, where each calibration input reveals only one action. The synthetic risk curves, four actions, five monotone bins, and  $\lambda = 0.18$  are the same as in Figure 2. The logging policy is deliberately cheap-biased:

$$\mu_\rho(0 \mid d) = 1 - 3\rho, \quad \mu_\rho(a \mid d) = \rho \quad \text{for } a \in \{1, 2, 3\}, \quad (108)$$

with known propensities and  $\rho \leq 1/4$ . For the logged action  $A_i$ , the observed objective is

$$Z_i = L_i + \lambda c_{A_i}, \quad L_i \sim \text{Bernoulli}(r_{A_i}(d_i)). \quad (109)$$

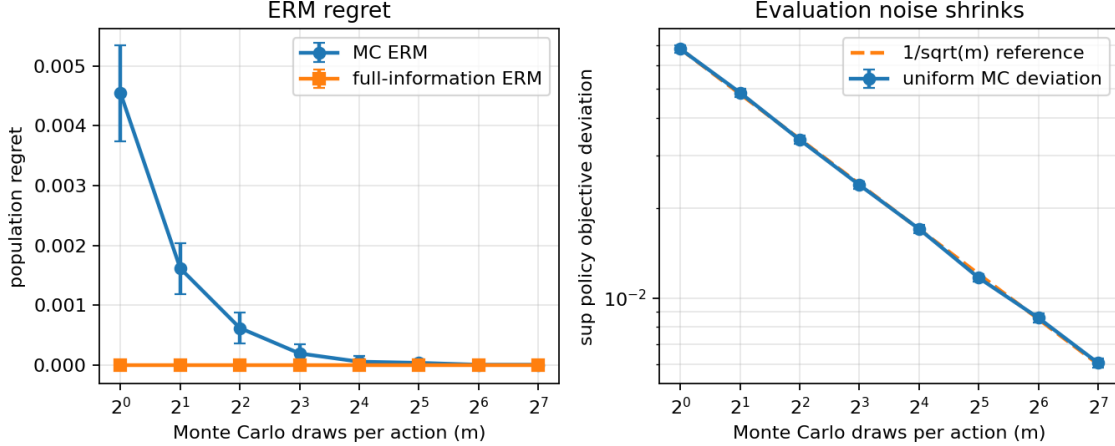


Figure 2: Direct PC-scale check of Corollary 5. Noisy bounded unbiased action evaluations induce excess regret and policy mismatches when  $m$  is small; the uniform Monte Carlo objective deviation follows the predicted  $1/\sqrt{m}$  trend.

IPS ERM minimizes  $n^{-1} \sum_i \mathbf{1}\{A_i = \pi(d_i)\} Z_i / \mu_\rho(A_i | d_i)$  over the same finite policy class and is evaluated on the deterministic population grid.

Figure 3 shows the cost of weak overlap. With  $n = 600$  logged examples and 400 repetitions, increasing  $\rho$  from 0.02 to 0.24 reduces the mean IPS policy regret from 0.0958 to 0.0058. The selected-policy mismatch rate relative to full-information ERM drops from 0.938 to 0.403, and  $\sup_{\pi \in \Pi} |\hat{J}^{\text{ps}}(\pi) - \hat{J}(\pi)|$  drops from 0.352 to 0.116. The finite-class Hoeffding bound again covers all repetitions but is very conservative, ranging from 9.46 at  $\rho = 0.02$  to 0.788 at  $\rho = 0.24$ . This experiment is a controlled check of known randomized logging and should not be read as covering deterministic logs, estimated propensities, or adaptive logging policies. The next check treats ordinary clipping separately because it changes the target objective.

## 10.4 Nested Sequential-Log IPS ERM Check

Early-exit and cascade systems often produce richer logs than ordinary one-action bandit feedback: running to a deeper action reveals all prefix outcomes. We ran a controlled experiment matching Corollary 7. The action menu, policy class, risk curves, and  $\lambda = 0.18$  are the same as in the logged-bandit check, but the behavior policy now draws a maximum reached depth  $A_i$ . It stops at depth zero with probability  $1 - 3\rho$  and at each deeper depth with probability  $\rho$ :

$$P(A_i = 0) = 1 - 3\rho, \quad P(A_i = a) = \rho \quad \text{for } a \in \{1, 2, 3\}. \quad (110)$$

Thus  $s_0 = 1$ ,  $s_1 = 3\rho$ ,  $s_2 = 2\rho$ , and  $s_3 = \rho$ . If  $A_i = a$ , the log reveals  $Z_{i,k} = L_{i,k} + \lambda c_k$  for every  $k \leq a$ . To make the comparison paired, each repetition first draws one full matrix of potential prefix losses and then censors it by  $A_i$ .

We compare two estimators on the same censored logs. The terminal-action IPS baseline keeps only examples with  $A_i = \pi(d_i)$  and divides by  $P(A_i = k)$ . The nested estimator keeps all examples with  $A_i \geq \pi(d_i)$  and divides by the survival probability  $P(A_i \geq k)$ . Thus the baseline discards deeper traces when evaluating shallower stopping policies, while the nested estimator uses them.

Figure 4 shows that prefix information materially changes policy selection. The plot includes two censoring-free references: a noisy full-information ERM that observes the same generated



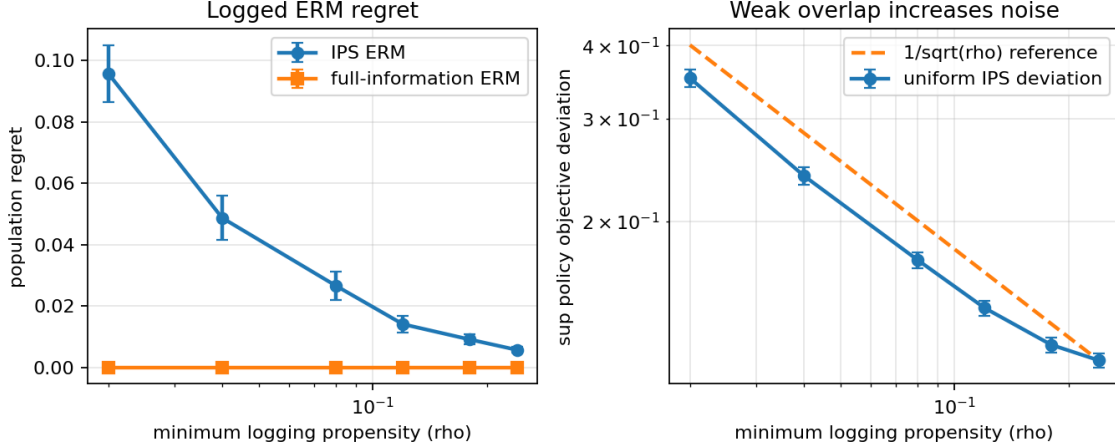


Figure 3: PC-scale logged-bandit check of Corollary 6. When the minimum logging propensity  $\rho$  is small, IPS ERM has larger objective deviation, more selected-policy mismatches, and higher population regret.

potential outcomes for all actions, and an expected-objective oracle. With  $n = 500$  and 400 repetitions, at  $\rho = 0.015$  the mean population regret is 0.1246 for terminal-action IPS and 0.0510 for nested survival IPS. At  $\rho = 0.22$ , the regrets are 0.00884 and 0.00211, respectively. The noisy full-information ERM has mean regret  $3.70 \times 10^{-4}$  in this configuration, so the gap between the logged estimators is not caused by full-information sampling noise. Uniform empirical objective deviation is also lower for nested IPS across all tested overlaps; for example, it drops from 0.441 to 0.276 at  $\rho = 0.015$  and from 0.125 to 0.0657 at  $\rho = 0.22$ . This is not a claim that nested logs remove overlap requirements. It shows that the relevant overlap is survival to the target stopping depth, not terminal stopping exactly at that depth.

## 10.5 Terminal-Propensity Separation Check

The previous nested-log check uses a logger whose smallest terminal and survival propensities are both controlled by the deepest action. To isolate Theorem 6, we also constructed a logger in which terminal stops at useful prefix actions are rare but survival to those prefixes is common. There are six ordered actions. The behavior logger stops at action 0 with probability 0.08, stops at each intermediate action  $1, \dots, 4$  with probability  $\epsilon$ , and otherwise runs to action 5:

$$P(A = 0) = 0.08, \quad P(A = k) = \epsilon \text{ for } k = 1, \dots, 4, \quad P(A = 5) = 0.92 - 4\epsilon. \quad (111)$$

Thus terminal-action IPS sees each intermediate prefix only with probability  $\epsilon$ , while nested IPS observes action  $k$  whenever  $A \geq k$ ; for  $\epsilon = 0.0025$ , the minimum terminal propensity over nonzero actions is 0.0025 but the minimum survival propensity is 0.91.

Figure 5 shows the resulting separation. With  $n = 600$  and 300 repetitions, terminal-action IPS has mean population regret 0.1548 at  $\epsilon = 0.0025$ , while nested survival IPS has regret 0.00594, close to the noisy full-information reference at 0.00543. The uniform objective deviation similarly drops from 1.18 for terminal-action IPS to 0.0437 for nested IPS. The usable logged fraction for the selected terminal policy is only 0.0041, while the selected nested policy is supported by a survival fraction 0.913. This experiment is a diagnostic, not a realistic benchmark: its purpose is to show

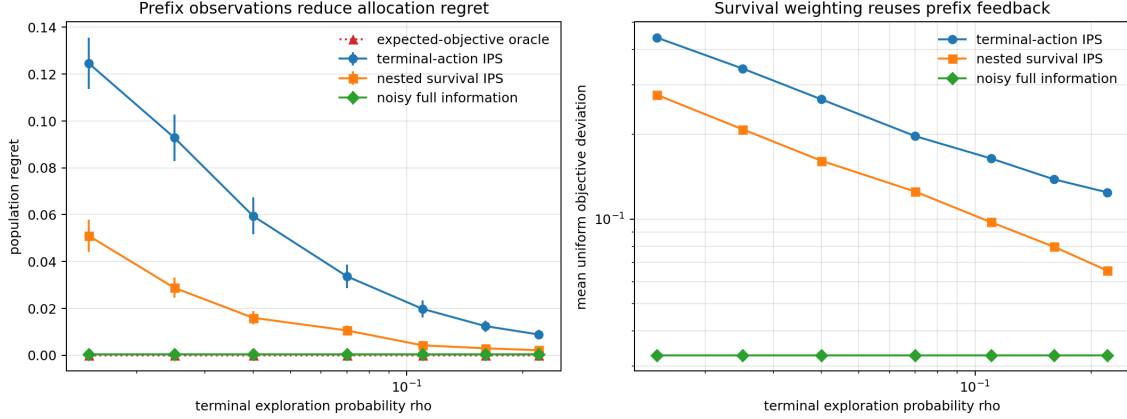


Figure 4: Nested sequential-log check of Corollary 7. When deeper traces reveal prefix objectives, survival-weighted IPS uses more logged information than terminal-action IPS. The noisy full-information and expected-objective references separate censoring error from ordinary action-evaluation noise.

that reducing nested traces to terminal actions discards information already present in the same runs.

We also varied the calibration sample size in the same terminal-propensity separation construction. Figure 6 reports the mean population regret over 40 repetitions, using sample sizes 150, 300, 600, 1200, 2400, and 4800, and terminal prefix probabilities 0.0025, 0.005, 0.01, 0.02, and 0.04. With target regret 0.04, nested survival IPS reaches the target at the smallest tested sample size ( $n = 150$ ) for every  $\epsilon$ . Terminal-action IPS does not reach the target by  $n = 4800$  for  $\epsilon \leq 0.01$ , reaches it at  $n = 2400$  for  $\epsilon = 0.02$ , and reaches it at  $n = 1200$  for  $\epsilon = 0.04$ . This is the finite-sample manifestation of the support parameter in Theorem 6: terminal reductions pay for rare exact prefix stops, while prefix-observing estimators pay for survival to the prefix.

## 10.6 Clipped IPS Bias–Variance Check

Finally, we ran a clipped-IPS diagnostic for Corollary 8. Here  $\rho = 0.02$  is fixed, so ordinary IPS corresponds to  $\tau = 1/\rho = 50$ . The logging policy is slightly more informative than in Figure 3: it assigns most probability to a difficulty-dependent base action and leaves probability  $\rho$  on each other action, so the highest-compute action is still observed only through exploration. All clipped estimates are computed on the same logged samples as ordinary IPS in each repetition.

Figure 7 shows the expected tradeoff. Small clip levels have low empirical deviation but large downward clipping bias; increasing  $\tau$  reduces the selected-policy bias and increases the empirical deviation. In this configured synthetic example, clipped IPS with  $\tau = 1$  has mean population regret 0.0121, selected-policy bias 0.266, and uniform empirical deviation 0.0363, while ordinary IPS at  $\tau = 50$  has zero clipping bias but regret 0.0439 and empirical deviation 0.331. The result should not be read as a universal improvement claim for clipping. It shows that, in finite logged calibration data, lower variance can select a better policy in some regimes, but the theoretical guarantee must pay for the induced bias.

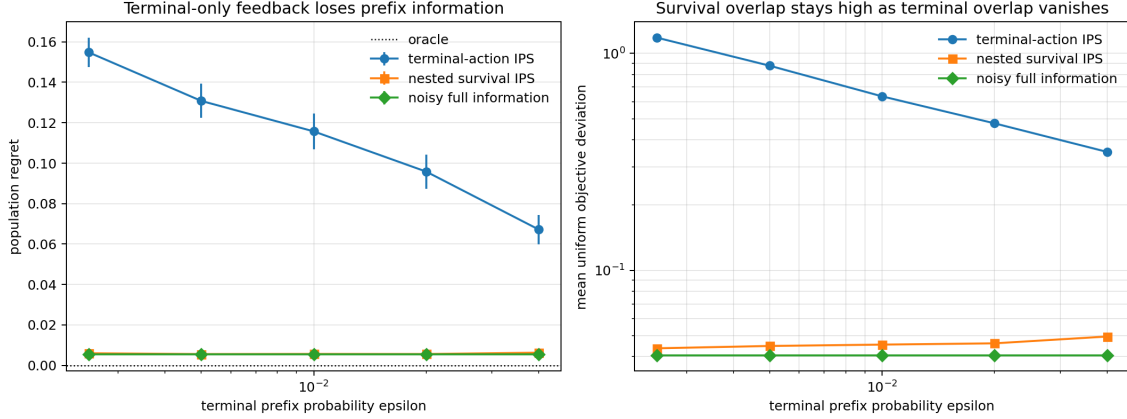


Figure 5: Terminal-propensity separation diagnostic for Theorem 6. Terminal-action IPS degrades as exact prefix stops become rare, while nested survival IPS remains close to noisy full-information ERM because deep traces reveal prefix outcomes.

## 10.7 Verifier Resampling Check

We also ran a verifier-resampling simulation using the same accept/reject process as Proposition 6. The plotted utility uses a reward/penalty version of the cost-augmented objective, assigning positive reward to true accepted candidates, penalty to false accepted candidates, and a fallback penalty when no candidate is accepted. The simulation sweeps the sample cap  $k$  under several verifier false-positive rates. Figure 8 shows that even small false-positive rates reduce the utility-maximizing sample cap; at larger false-positive rates, long resampling runs increase the probability that the first accepted candidate is incorrect and become inferior once sample cost is included. This simulation visualizes the same verifier failure mode, but it does not plot exactly the  $P_k$ ,  $U_k$ , or  $\tilde{U}_k$  quantities in the proposition. It is also not an empirical instance of Corollary 5, because we do not perform calibration ERM from noisy action estimates and then evaluate the selected policy on fresh deployment data.

We also ran the latent-difficulty calculation from Proposition 7. All curves in Figure 9 have the same mean per-attempt success probability,  $\mathbb{E}[\theta] = 0.08$ . The homogeneous curve uses a constant  $\theta$ , the beta curves increase the variance of  $\theta$  without an atom at zero, and the zero-inflated curves assign a positive fraction of inputs to  $\theta = 0$ . The result illustrates two separate effects: heterogeneity makes the homogeneous iid extrapolation optimistic at finite  $k$ , and an atom at zero creates a ceiling because some inputs are unsolved by this sampling mechanism. This is an analytic distributional check, not a Monte Carlo action-evaluation experiment.

## 10.8 Text-Classifer Routing Proxy

As a non-synthetic routing proxy, we also ran the fixed-action calibration workflow on the full 20 Newsgroups text classification task [60]. The action menu consists of three TF-IDF Multinomial Naive Bayes classifiers with increasing feature budgets: 1,000 unigram features, 6,000 unigram features, and 20,000 unigram/bigram features. A cheap classifier supplies the confidence score used to define five train-split quantile bins. For each  $\lambda$ , calibration selects a monotone confidence-bin assignment either from full-information held-out losses or from simulated known-propensity terminal logs with  $\rho = 0.08$ . Deployment cost is measured warm-batch CPU prediction latency per test example; adaptive policies pay the cheap router cost before any larger selected action, while

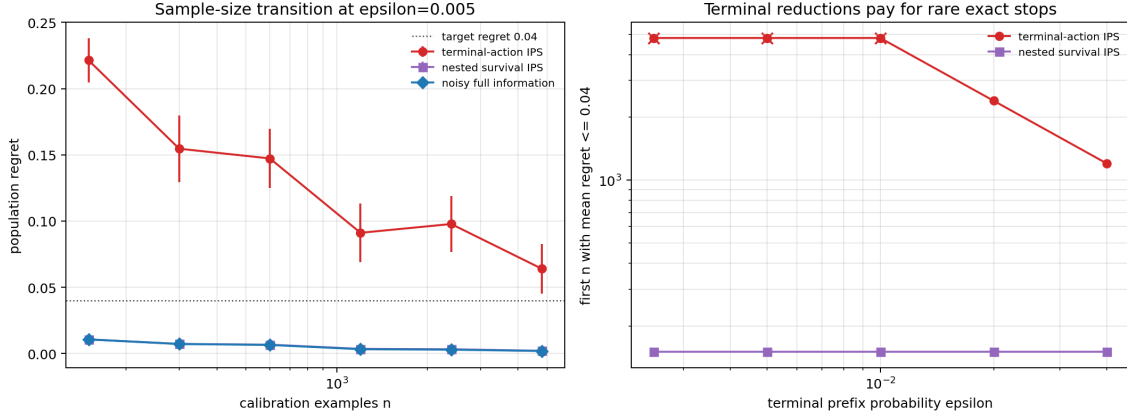


Figure 6: Sample-size diagnostic for the terminal-reduction separation. Left: regret as calibration size grows at  $\epsilon = 0.005$ . Right: first tested calibration size at which mean regret is at most 0.04; crosses mark terminal-action settings that remain above target at the largest tested sample size.

fixed baselines are measured standalone.

Figure 10 shows the resulting accuracy-latency tradeoff over ten stratified repetitions. The fixed small, medium, and large actions obtain test accuracies 0.519, 0.648, and 0.678, respectively. Full-information binned routing moves along the same frontier as  $\lambda$  increases: from accuracy 0.678 at 0.184 ms per example for  $\lambda = 0$  to accuracy 0.670 at 0.149 ms for  $\lambda = 0.10$ . The logged-IPS router is less stable, as expected under one-action feedback: at  $\lambda = 0.10$  it obtains accuracy 0.654 at 0.136 ms.

The label-aware oracle is shown only as an upper reference. This experiment is not intended as a benchmark claim about state-of-the-art routing. Its role is construct validity: the same fixed action menu, cheap routing signal, held-out calibration split, measured deployment cost, and cost-augmented objective appear in a real text classification pipeline.

## 10.9 Ordered Text Routing and Prefix Feedback

We also ran an ordered variant of the same text-routing proxy to instantiate the nested-log distinction on non-synthetic data. The three classifiers are treated as stopping depths: action 0 runs only the small classifier, action 1 runs small then medium, and action 2 runs all three. A deeper run therefore reveals the prefix losses for shallower classifiers. Costs are cumulative measured CPU latencies. The max-depth logger has  $P(A = 0) = 0.05$ ,  $P(A = 1) = \epsilon$ , and  $P(A = 2) = 0.95 - \epsilon$ , so survival to the medium prefix is 0.95 even when terminal stops at that prefix are rare.

Figure 11 shows the result. At  $\lambda = 0.10$  and  $\epsilon = 0.0025$ , terminal-action IPS has mean test risk gap 0.00356 relative to full-information binned routing and changes the selected action on 38.4% of test examples. Terminal self-normalized IPS (SNIPS) and terminal clipped IPS have gaps 0.00366 and 0.00489. Nested survival IPS selects the same policies as the full-information binned reference in these ten repeats, with mean test risk gap 0.00000. The selected weight is 248.3 for terminal IPS; for the nested-selected policy, it is 1.04. This does not make a large-model routing benchmark claim; it shows that the survival-vs-terminal feedback distinction appears in the same kind of fixed-action, measured-latency routing workflow used above.

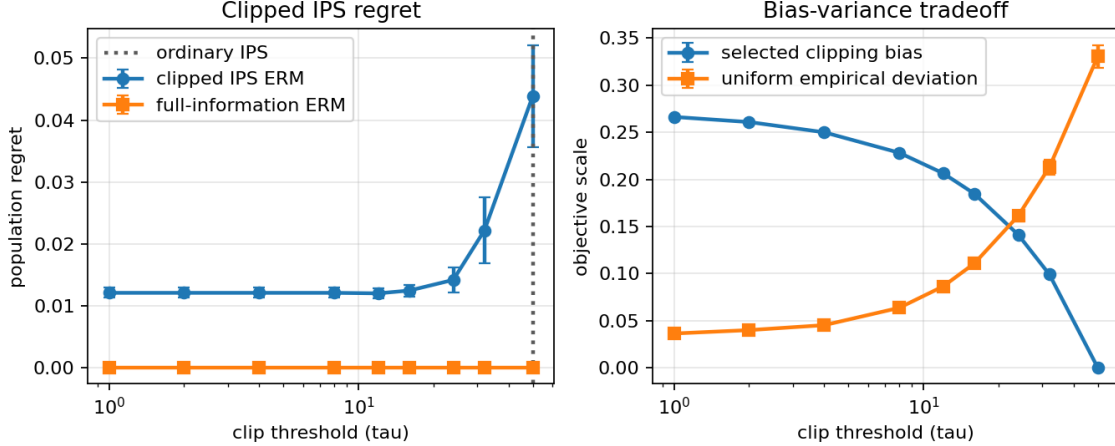


Figure 7: PC-scale clipped IPS diagnostic for Corollary 8. Reducing the clip threshold lowers empirical objective deviation but introduces clipping bias relative to the true deployment objective.

### 10.10 Sampled-Answer Trace Proxy

To move closer to self-consistency style test-time compute while staying PC-only, we also constructed cached sampled-answer traces on 20 Newsgroups. A fixed probabilistic text classifier produces a distribution over labels. For each document we cache 16 sampled candidate labels, and the action menu  $k \in \{1, 2, 4, 8, 16\}$  returns the majority vote among the first  $k$  samples. Running to a larger  $k$  therefore records every smaller-prefix majority vote and loss. The max-sample logger has  $P(A = 1) = 0.05$ , assigns probability  $\epsilon$  to each intermediate sample count, and places the remaining mass on  $A = 16$ .

Figure 12 shows the paired comparison. At  $\lambda = 0.08$  and  $\epsilon = 0.0025$ , terminal IPS, SNIPS, and clipped IPS have mean risk gaps 0.0210, 0.0210, and 0.0217 relative to full-information binned routing and change the selected sample count on 76.1%, 74.7%, and 77.4% of test examples. Nested survival IPS has mean risk gap 0.00039 and changes 3.4% of actions. The selected exact-match fraction for the terminal IPS-selected policy is about 0.052, with average selected weight 335.9; the selected survival fraction for the nested-selected policy is about 0.953, with average selected weight 1.05. This is not a language-model benchmark. It is a cached trace check that the same max-sample logs should be analyzed by prefix survival rather than compressed to terminal pairs.

Finally, we made the ordered text logger input-dependent. The max-depth logger uses the cheap classifier confidence bin: easy high-confidence examples are more likely to stop at the small classifier, while harder examples are more likely to run deep. This keeps propensities known but makes the correct survival probability contextual. Figure 13 compares four estimators at  $\lambda = 0.10$  and  $\epsilon = 0.0025$ : terminal IPS with contextual terminal probabilities, nested IPS with contextual survival probabilities, and two ablations that replace those contextual probabilities by their marginal averages. As logger dependence increases to 0.45, contextual nested survival IPS remains close to full-information routing, with mean risk gap 0.00019, whereas marginal survival drifts to a gap of 0.00332 and changes 23.2% of actions. Terminal estimators have larger gaps because exact medium-prefix stops are rare. This diagnostic supports the identification message: survival is the right support event for prefix outcomes, but the survival probability must be conditioned on the logging context when the continuation mechanism is contextual.

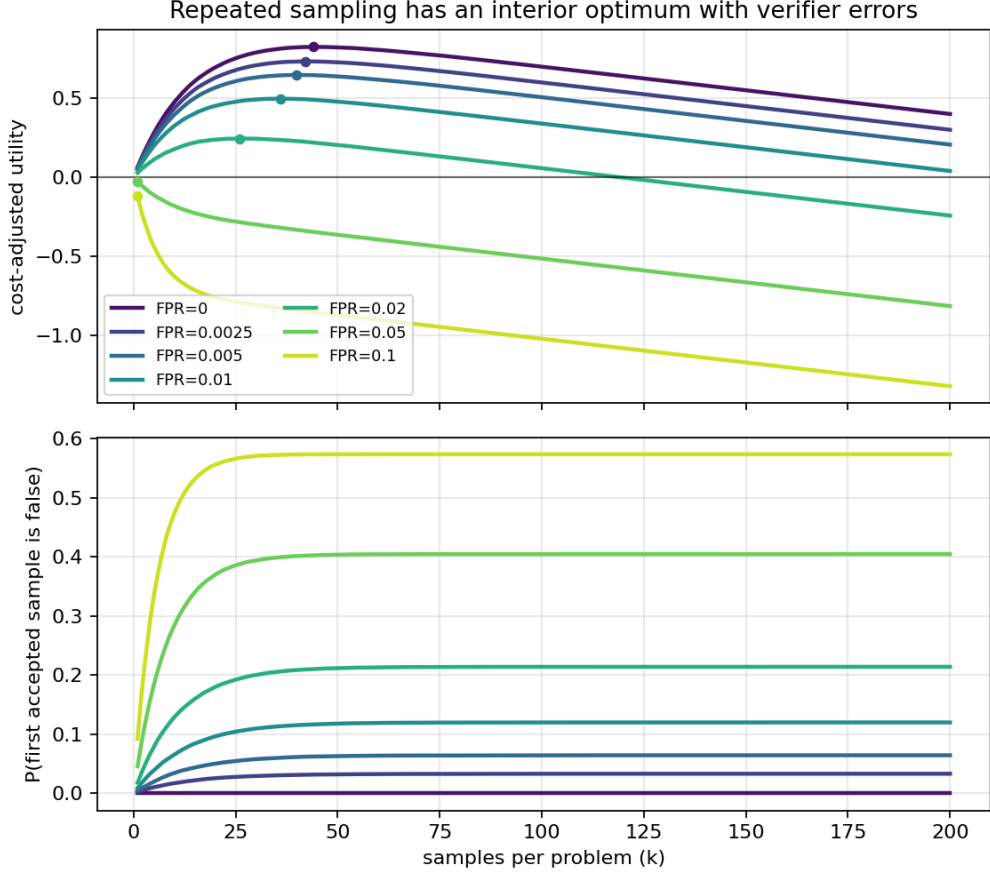


Figure 8: PC-scale simulation of verifier-guided resampling. As verifier false positives increase, the utility-optimal sample cap decreases and the probability that the first accepted candidate is incorrect grows with  $k$ .

### 10.11 Staged Boosting Trace Proxy

As a second prefix-observing trace diagnostic, we used a single Gradient Boosting classifier on scikit-learn digits and treated staged predictions as ordered compute actions. The checkpoints are 5, 10, 20, 40, 80, and 120 boosting stages. Running to a later checkpoint records the predictions and losses for every earlier checkpoint, so the same max-checkpoint log can again be analyzed either as terminal-action feedback or as a prefix-observing trace. The deployment cost is the checkpoint index normalized by the final checkpoint.

Figure 14 shows the paired comparison over 20 repeats. At  $\lambda = 0.05$  and  $\epsilon = 0.005$ , terminal IPS has mean test risk gap 0.0387 relative to full-information binned routing and changes 89.8% of selected checkpoints. Nested survival IPS has mean gap  $-0.00029$  and changes 7.7% of checkpoints. The selected exact-match fraction for the terminal-selected policy is 0.00181, with average selected weight 192; the selected survival fraction for the nested-selected policy is 0.957, with average selected weight 1.05. This does not claim that boosting stages are a modern large-model workload. It checks the same observation-operator boundary in a single progressive model rather than a menu of separately trained classifiers.

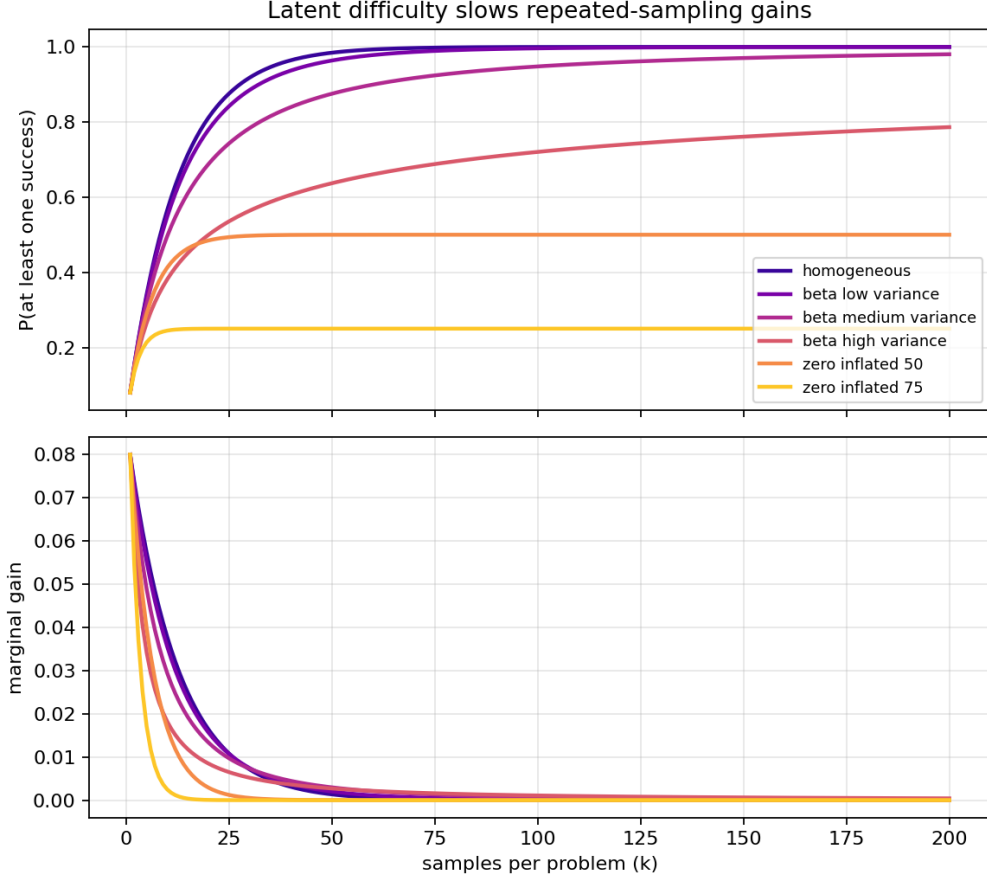


Figure 9: Analytic PC-scale check for latent input difficulty in repeated sampling. All scenarios have the same average per-attempt success probability, but heterogeneous  $\theta$  slows best-of- $k$  gains and zero-inflated  $\theta$  creates an unsolved-input ceiling.

### 10.12 Small Real-Data Case Study: Digits

To complement the synthetic checks, we instantiated the same finite-action objective on scikit-learn’s handwritten digits dataset [61, 62]. The actions are random-forest classifiers [63] using 16, 32, or 64 train-set variance-ranked pixels. A cheap 16-pixel classifier provides a confidence-based difficulty proxy. Train-set confidence quantiles fix five difficulty bins before calibration, and for each  $\lambda$  the procedure selects a monotone bin-to-action assignment minimizing empirical 0-1 loss plus  $\lambda$  times normalized proxy cost. The normalized costs are 0.20, 0.60, and 1.00, because the cost model includes the cheap router before optional larger actions, including for fixed baselines. We repeat stratified 60/20/20 train/calibration/test splits 30 times and report mean performance with 1.96 standard-error bars in Figure 15. This setup instantiates the fixed-binning restricted-assignment case: conditional on the training split, calibration selects only among monotone bin-to-action assignments. It does not evaluate Corollary 10, which would require calibration to choose among multiple candidate binnings. The calibration losses are exact full-information 0-1 losses for deterministic trained classifiers and fixed proxy costs, so Corollary 5 is not invoked.

The fixed 16-, 32-, and 64-pixel actions obtain mean test accuracies 0.937, 0.960, and 0.964 at normalized costs 0.20, 0.60, and 1.00, respectively. The adaptive binned policy moves along a



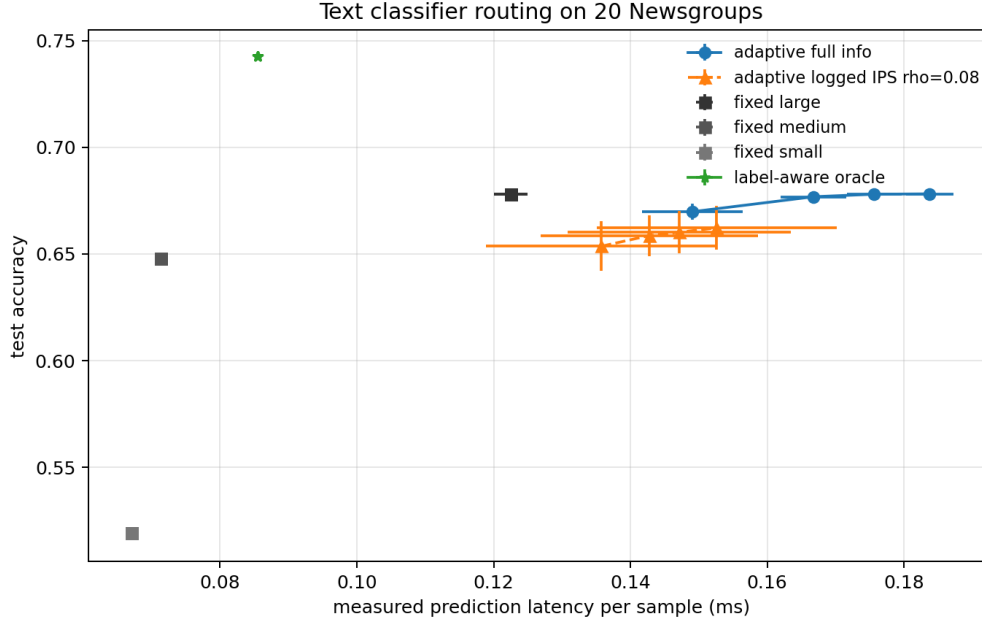


Figure 10: CPU-scale text-classifier routing proxy on 20 Newsgroups. The adaptive policies use cheap confidence bins and measured latency costs; they are construct-validity checks, not large-language-model routing benchmarks.

cost-accuracy curve as  $\lambda$  increases: for example,  $\lambda = 0.02$  gives accuracy 0.960 at cost 0.428, while  $\lambda = 0.10$  gives accuracy 0.957 at cost 0.348. The full-information oracle reaches accuracy 0.979 at cost 0.220, but it is a label-aware upper reference rather than a deployable policy. This case study is therefore not evidence that the simple router is empirically optimal; it shows that the policy objective, calibration split, and budget tradeoff can be reproduced on a small real dataset.

We additionally ran a latency-measured variant of the digits experiment, shown in Figure 16. The action family uses random forests with increasing feature counts and tree counts: 16 pixels with 40 trees, 32 pixels with 80 trees, and 64 pixels with 160 trees. Across 30 stratified splits, we measured warm batch per-sample CPU prediction time on the local machine, using two warm-up passes and the median of 20 timed passes. The confidence-bin edges and latency normalizer used by the calibration objective are fixed from the train split before calibration. Adaptive policies pay the cheap 16-pixel router cost before any larger selected action; fixed-action baselines are measured standalone and should be read as external reference points rather than policies in the same routed action class. The fixed small, medium, and large actions obtain accuracies 0.920, 0.960, and 0.965 at mean measured latencies 0.0106, 0.0208, and 0.0406 ms per sample. The adaptive policy trades accuracy for measured latency as  $\lambda$  increases:  $\lambda = 0.02$  gives accuracy 0.963 at 0.0252 ms,  $\lambda = 0.05$  gives accuracy 0.961 at 0.0214 ms, and  $\lambda = 0.10$  gives accuracy 0.959 at 0.0188 ms. The raw latency CSV stores the split seeds, timing repeats, software versions, and platform/machine strings. These timings are a local systems sanity check for using an empirical cost term, not a hardware-independent speedup claim. The timing variability is a systems measurement issue; the reported median warm-batch timings are treated as empirical costs rather than bounded unbiased Monte Carlo estimates of per-example action objectives.



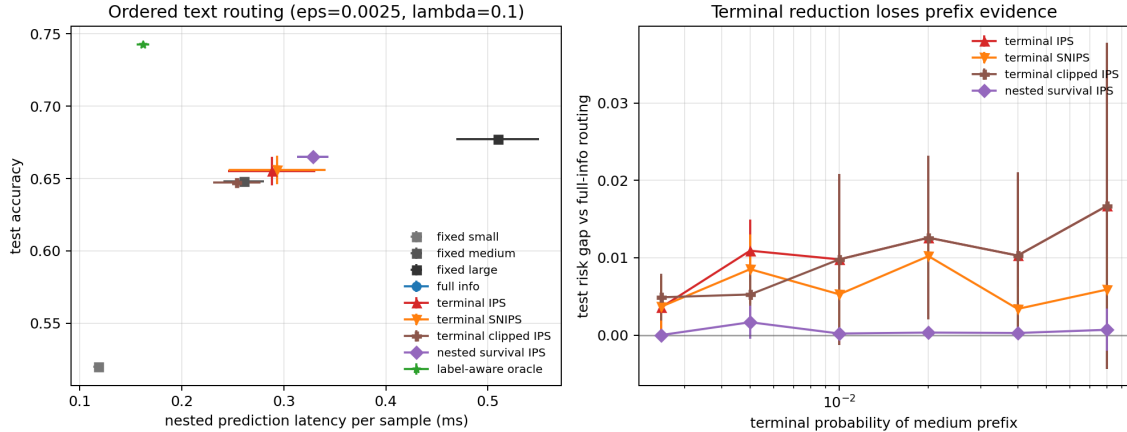


Figure 11: Ordered 20 Newsgroups routing proxy. When deeper classifier runs reveal prefix outcomes, survival-weighted nested IPS stays close to full-information routing while terminal-action IPS suffers from rare exact terminal stops at the medium prefix.

Table 3: Selected digits case-study results. Proxy costs are normalized feature-budget costs; latency costs are warm CPU milliseconds per sample. Parenthesized values are 1.96 standard errors.

| Case    | Policy          | $\lambda$ | Accuracy      | Cost or latency    |
|---------|-----------------|-----------|---------------|--------------------|
| Proxy   | uniform 16px    | 0         | 0.937 (0.005) | 0.200 (0.000)      |
| Proxy   | uniform 32px    | 0         | 0.960 (0.003) | 0.600 (0.000)      |
| Proxy   | uniform 64px    | 0         | 0.964 (0.003) | 1.000 (0.000)      |
| Proxy   | adaptive binned | 0.02      | 0.960 (0.003) | 0.428 (0.030)      |
| Proxy   | adaptive binned | 0.10      | 0.957 (0.003) | 0.348 (0.018)      |
| Latency | uniform small   | 0         | 0.920 (0.006) | 0.0106 (0.0009) ms |
| Latency | uniform medium  | 0         | 0.960 (0.004) | 0.0208 (0.0015) ms |
| Latency | uniform large   | 0         | 0.965 (0.004) | 0.0406 (0.0026) ms |
| Latency | adaptive binned | 0.02      | 0.963 (0.004) | 0.0252 (0.0025) ms |
| Latency | adaptive binned | 0.10      | 0.959 (0.004) | 0.0188 (0.0016) ms |

## 11 Limitations

The paper treats the available compute actions as fixed. It analyzes how to choose among them and how estimation errors propagate to deployment risk. It does not design stronger base predictors, stronger verifiers, better search procedures, or new routing architectures. In particular, learning the conditional objectives  $q_a(x)$  for large generative models remains a separate and difficult problem.

The feedback models are deliberately clean. Full-information calibration, bounded independent Monte Carlo action evaluation, known-propensity randomized logging, and known-propensity clipped IPS are useful reference cases, but they do not cover biased verifier scores, estimated propensities, self-normalized or doubly robust theoretical guarantees, hidden confounding, covariate shift, or arbitrary hardware timing noise. For nested adaptive stopping logs, marginal survival propensities are insufficient when continuation depends on observed verifier scores, confidences, or intermediate losses; known history-dependent sequential propensities are then needed. If those continuation variables are unobserved by the analyst, the problem becomes a hidden-confounding

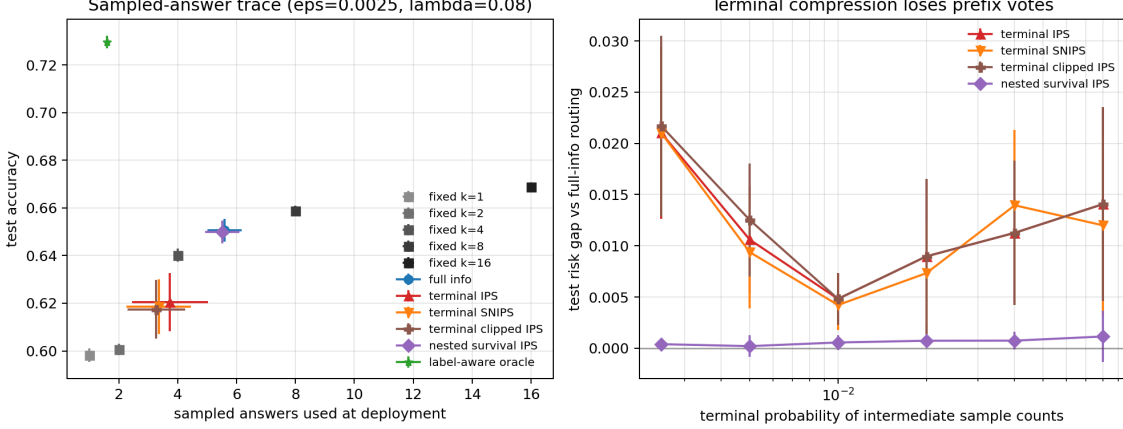


Figure 12: Cached sampled-answer trace on 20 Newsgroups. Actions are majority votes over prefixes of sampled labels. Terminal-action IPS discards prefix votes revealed by deeper sampled-answer traces; nested survival IPS tracks full-information routing on the same logs.

problem outside our guarantees. The prefix result applies only to retained trace fields that identify the prefix objective; if a production log stores only the terminal answer, or if continuing computation changes the prefix object that would have been observed under stopping, the data reduce to ordinary terminal or history-dependent censored feedback. The clipped-IPS guarantee is an explicit-bias statement, not a claim that clipping preserves the original deployment objective.

Inference cost is deployment-specific. FLOPs, tokens, latency, energy, and money can induce different orderings of the same actions, and measured latency depends on hardware, batching, implementation, and load. The theory requires a chosen cost function; it does not decide which cost is appropriate for a particular product or system.

The empirical section is intentionally small. The synthetic experiments instantiate assumptions exactly, and the digits experiments use classical random forests on a low-dimensional dataset. They check the formalism on a personal computer, but they are not evidence of production speedups, LLM routing quality, or high-dimensional representation learning behavior. Finite-class and PAC-Bayes bounds may also be numerically loose for modern deep systems; their role here is to identify the allocation quantities that sharper analysis should control.

## 12 Conclusion

Adaptive inference makes computation an input-dependent decision. This paper’s central claim is that, after the available compute actions are fixed, the key object is the conditional action objective  $q_a(x) = r_a(x) + \lambda c_a(x)$ . Estimating this objective well is enough to control deployment regret: a uniform objective error  $\eta$  gives a sharp  $2\eta$  plug-in allocation bound, and margin localization shows that errors matter primarily near action-tie boundaries. For fixed difficulty bins, the same localization survives finite calibration: nonempty bins with separated best actions drop out of the regret bound once their confidence intervals are narrow enough.

The same view organizes finite-sample calibration, average-budget relaxations, ordered stopping, logged feedback, clipping bias, and simple resampling ceilings. The contribution is therefore not that cost-sensitive learning is new, but that modern adaptive inference can be reduced to a precise action-objective estimation problem with transparent statistical and feedback assumptions. The

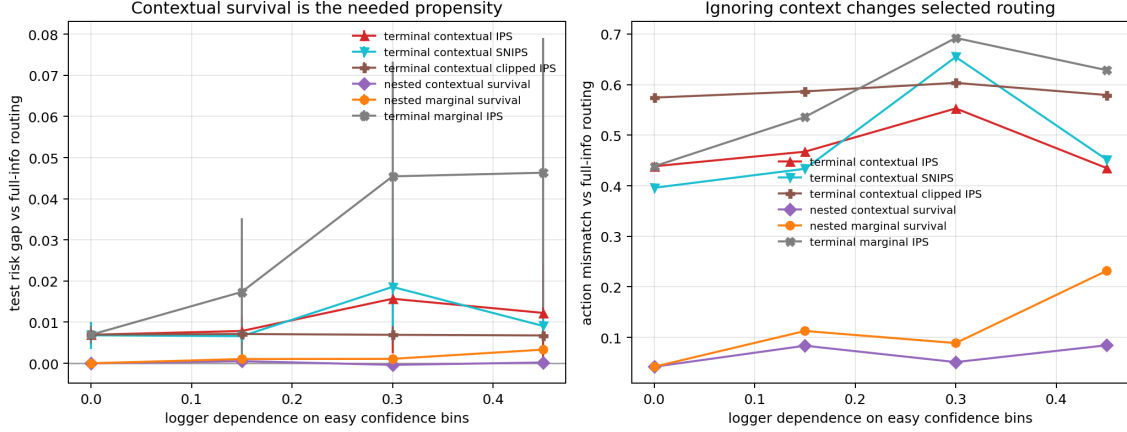


Figure 13: Contextual ordered text-routing diagnostic. The max-depth logger depends on cheap confidence bins. Correct contextual survival weighting stays close to full-information routing, while marginal-propensity ablations drift as the logger becomes more input-dependent.

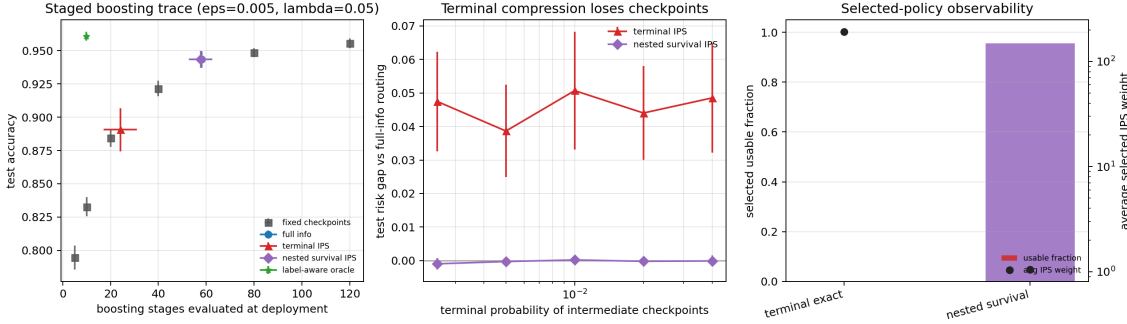


Figure 14: Progressive staged-inference trace on sklearn digits. A single Gradient Boosting model exposes prefix checkpoint predictions; terminal compression discards checkpoint losses already observed by deeper runs, while nested survival IPS tracks full-information routing.

included experiments are reproducible PC-scale checks of this theory; stronger empirical claims require domain-specific action sets, costs, and deployment data.

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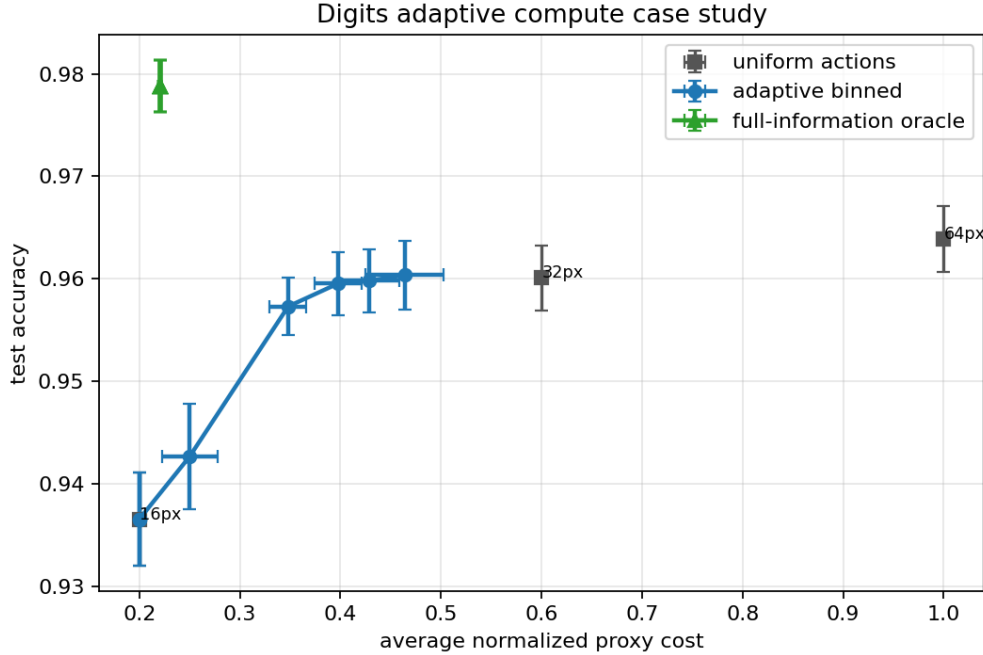


Figure 15: Small real-data case study on handwritten digits. Costs are normalized feature-budget proxies, not measured latency or FLOPs. The adaptive curve illustrates cost-sensitive action selection but does not dominate all fixed baselines.

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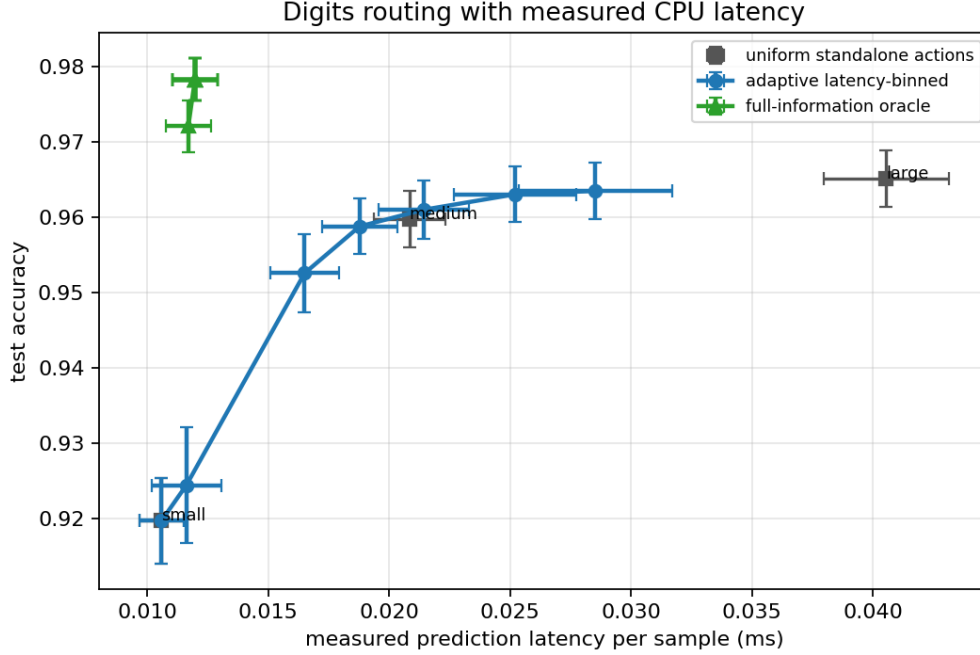


Figure 16: Measured-latency variant of the digits case study. Costs are warm batch per-sample CPU prediction latencies on the local machine. Adaptive policies include the cheap-router cost; fixed baselines do not. Absolute timings are hardware-, implementation-, and batch-size-dependent.

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